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Darekar et al.

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(54) **SKIRT-REINFORCEMENT MEMBERS FOR PROSTHETIC VALVE DEVICES**

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Related U.S. Application Data

(60) Provisional application No. 63/129,166, filed on Dec.
22, 2020.

(51) **Int. Cl.**
A61F 2/24 (2006.01)

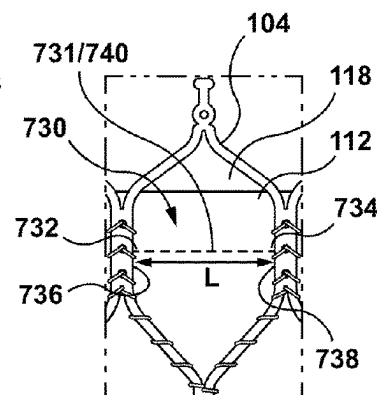
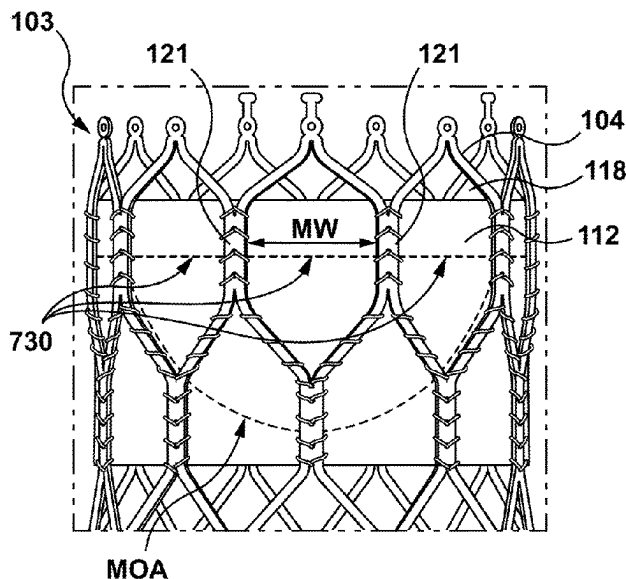
(52) **U.S. Cl.**
CPC **A61F 2/2409** (2013.01); **A61F 2/2418**
(2013.01); **A61F 2220/0075** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61F 2/24; A61F 2/2409; A61F 2/2418;
A61F 2/2457; A61F 2/2412;
(Continued)

(57) **ABSTRACT**

A prosthesis includes a frame and a skirt coupled to a surface of the frame. The skirt extends over at least one side opening of a plurality of side openings of the frame. The prosthesis also includes a skirt-reinforcement member that spans the at least one side opening from a first crown or strut edge defining the at least one side opening to a second crown or strut edge defining the at least one side opening. The skirt-reinforcement member has a first end, a second end, and a length therebetween. The first end of the skirt-reinforcement member is attached to the first crown or strut edge and the second end of the skirt-reinforcement member is attached to the second crown or strut edge. The skirt-reinforcement member is attached to the skirt along an unsupported portion of the skirt that spans the at least one side opening.

20 Claims, 18 Drawing Sheets



(52) **U.S. Cl.**

CPC . A61F 2230/0067 (2013.01); A61F 2250/001
(2013.01); A61F 2250/0039 (2013.01)

(58) **Field of Classification Search**

CPC A61F 2250/0063; A61F 2/852; A61F 2/82;
A61F 2/2442; A61F 2/2454

See application file for complete search history.

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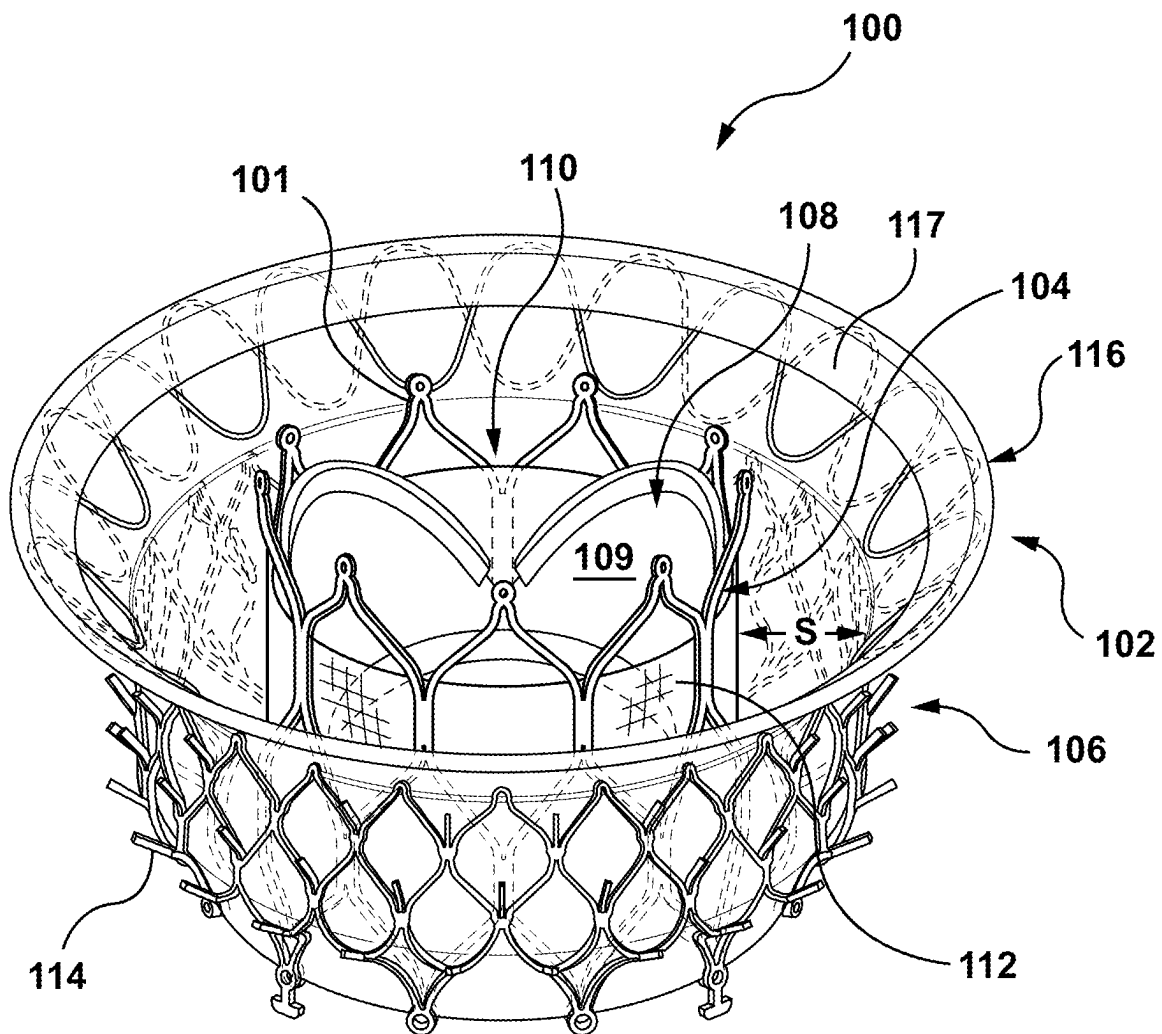


FIG. 1

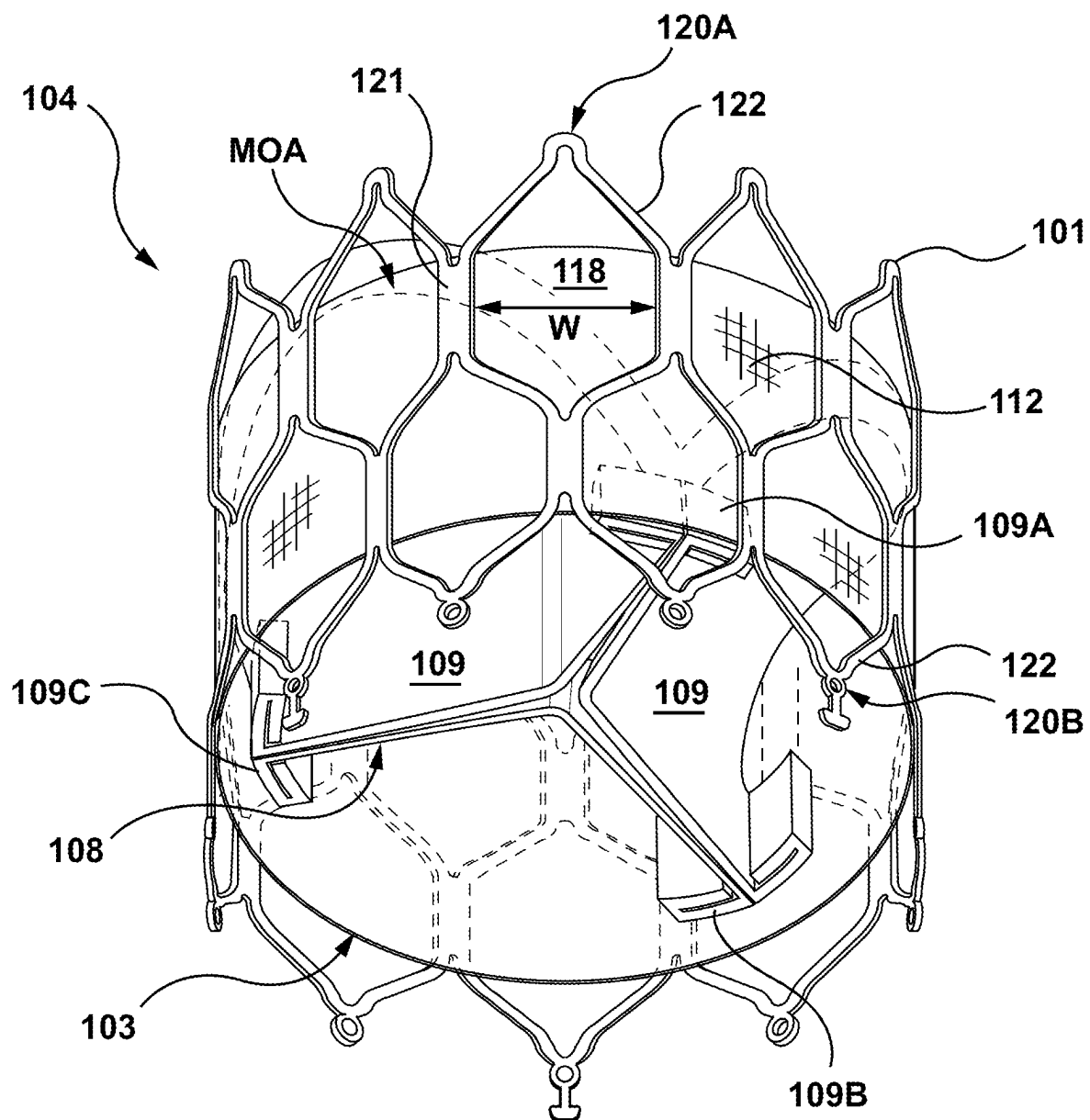


FIG. 2

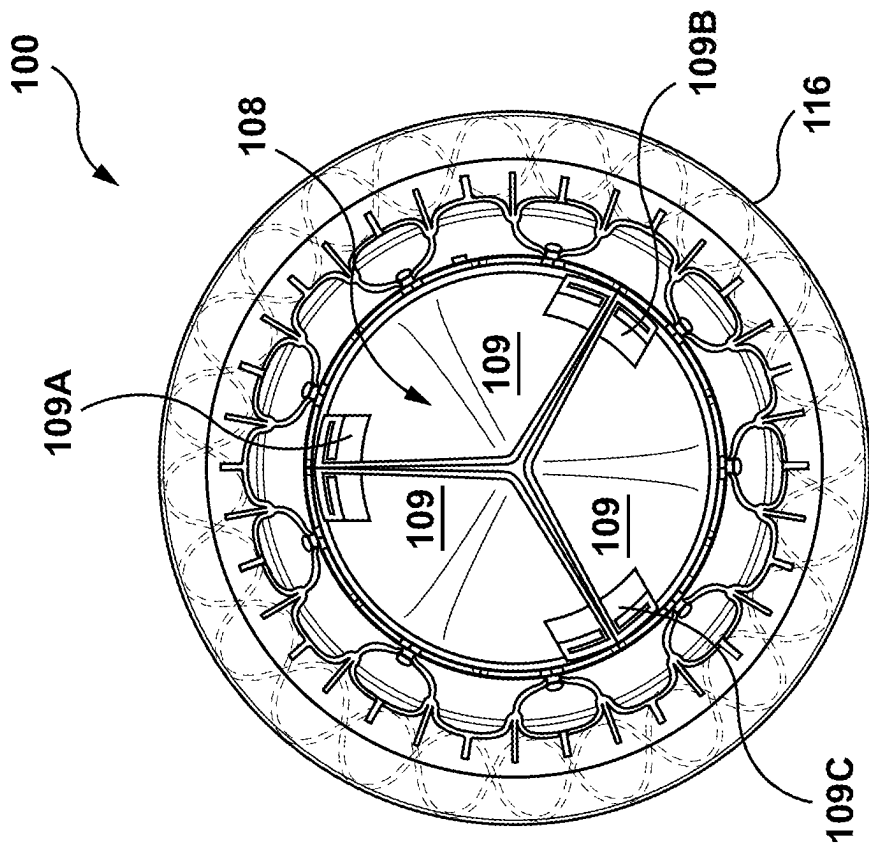


FIG. 4

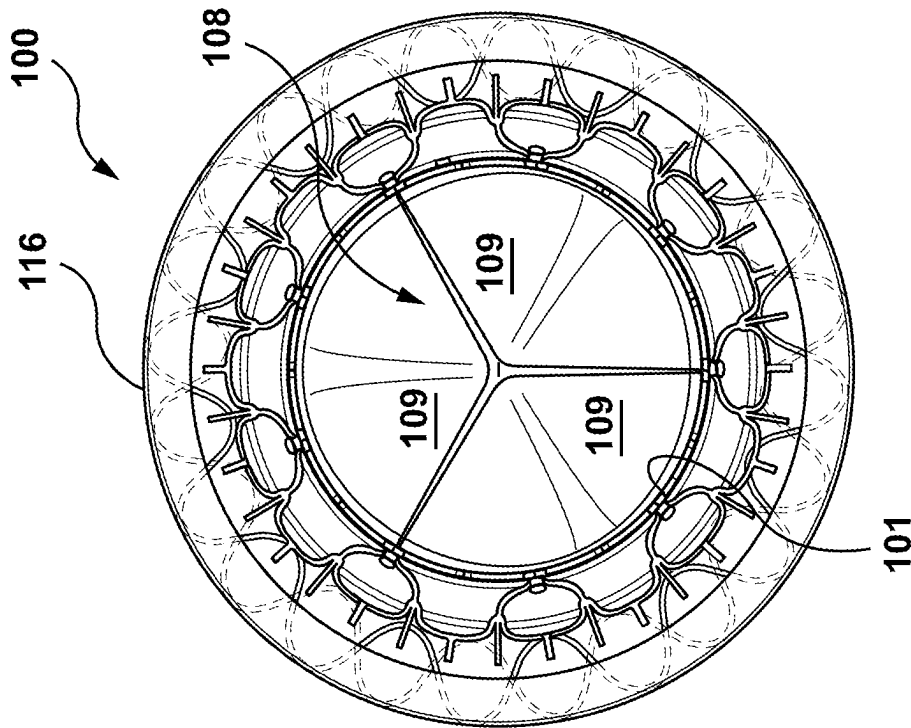


FIG. 3

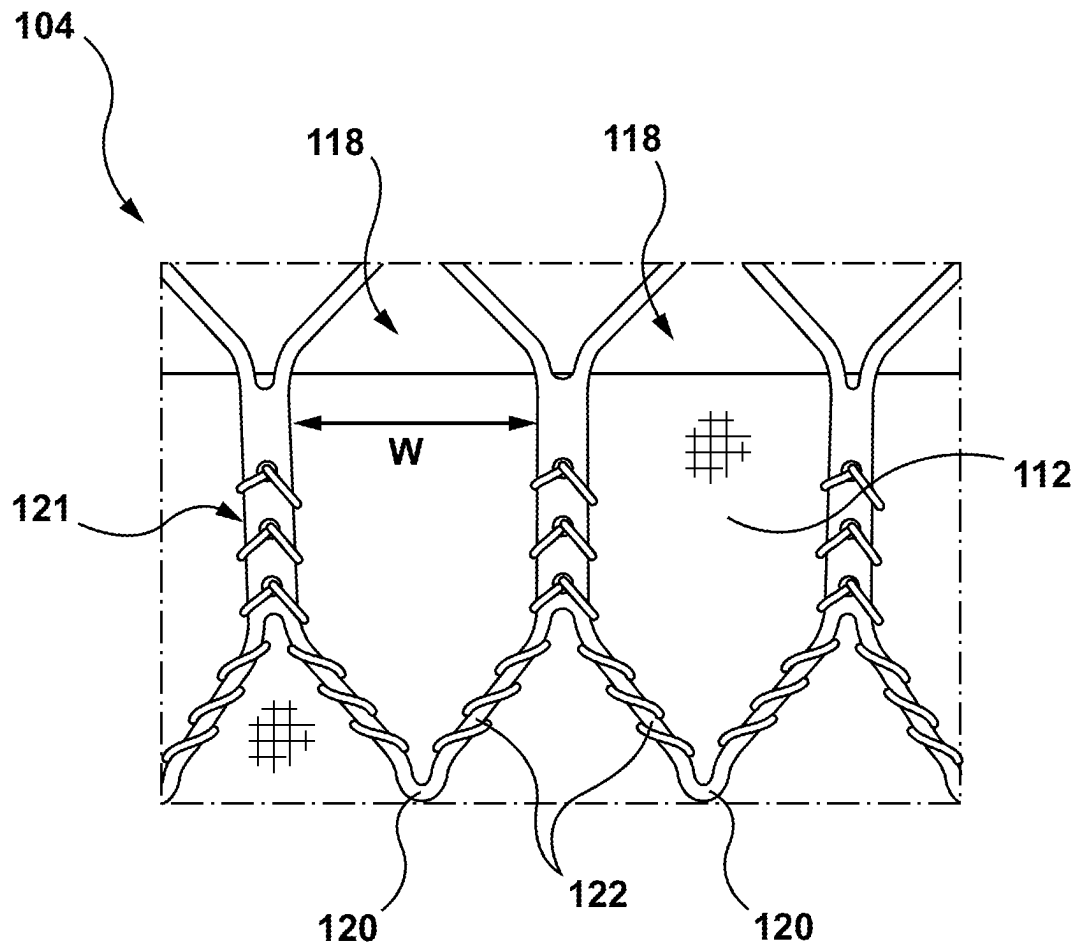


FIG. 5

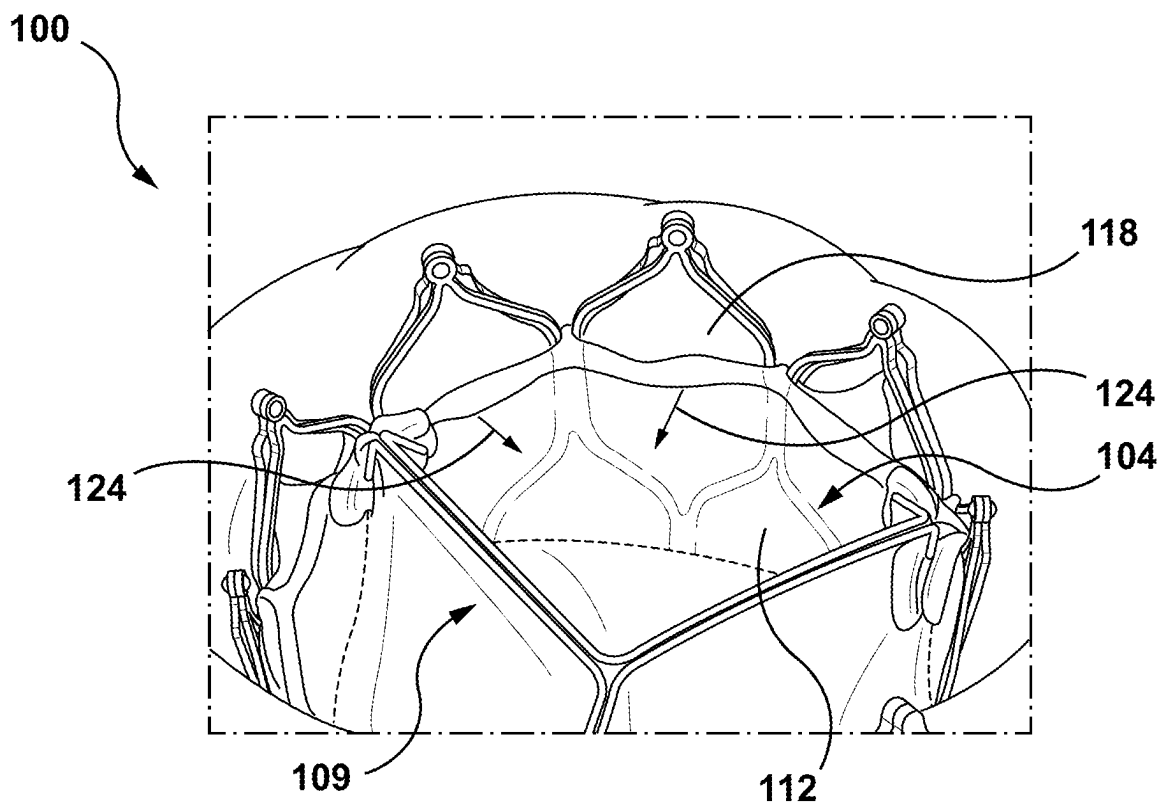


FIG. 6A

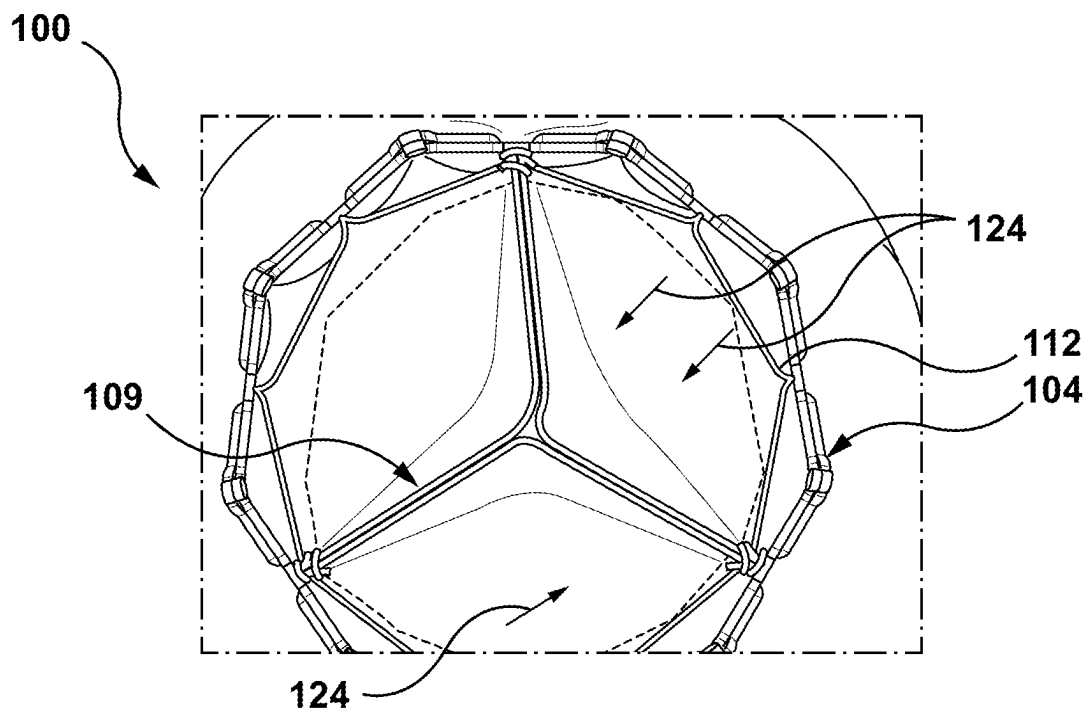
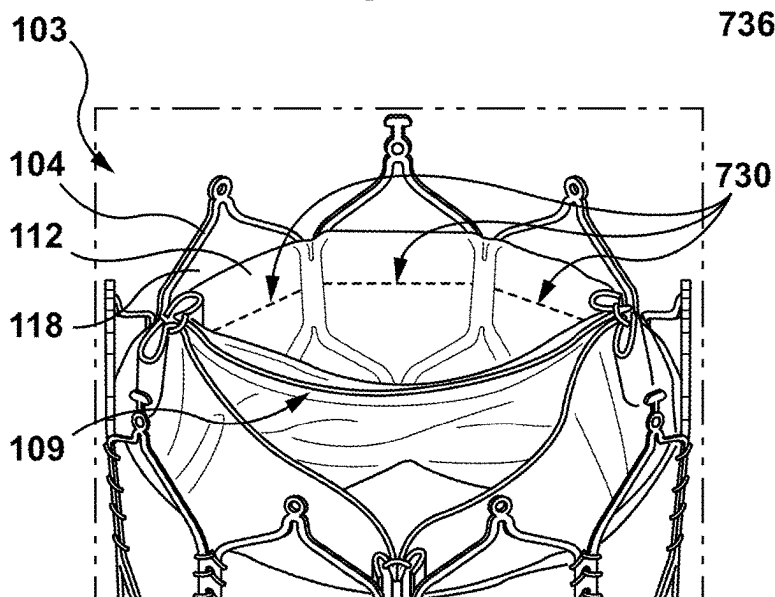
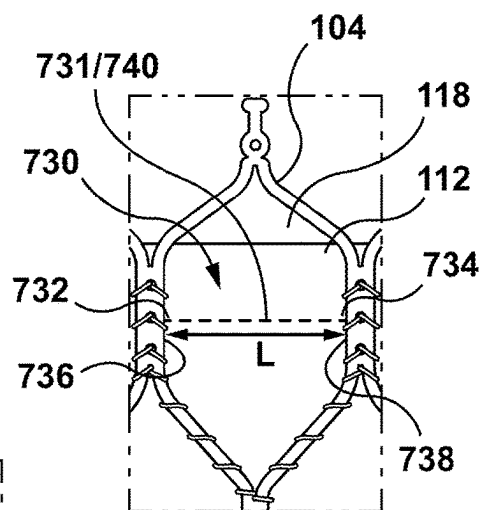
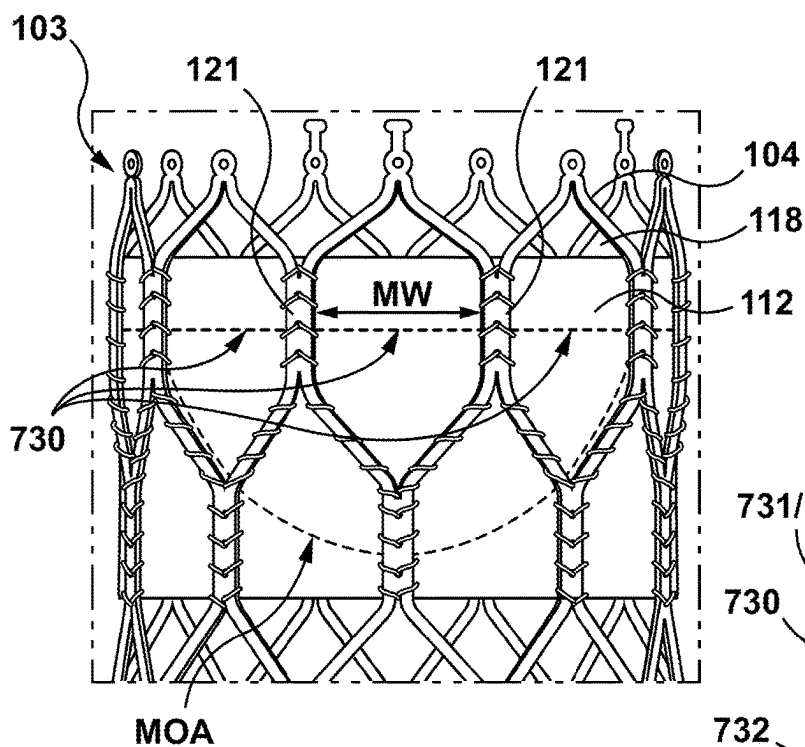
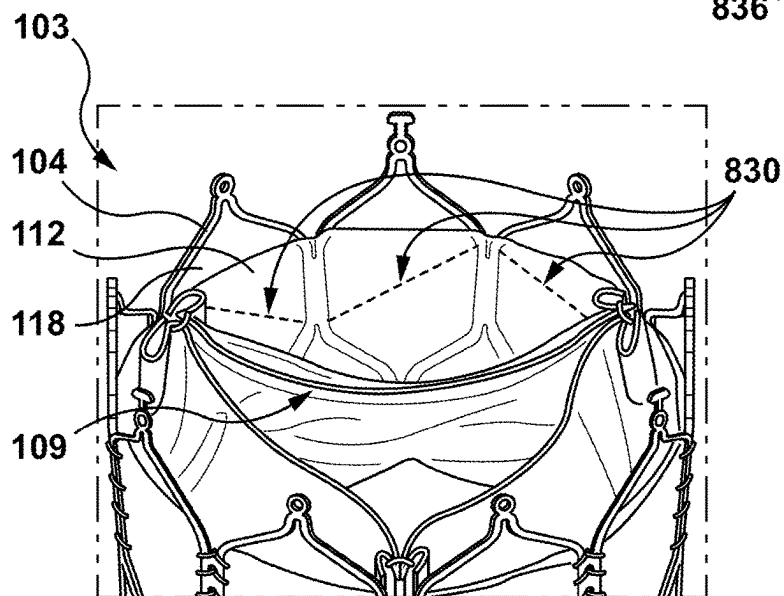
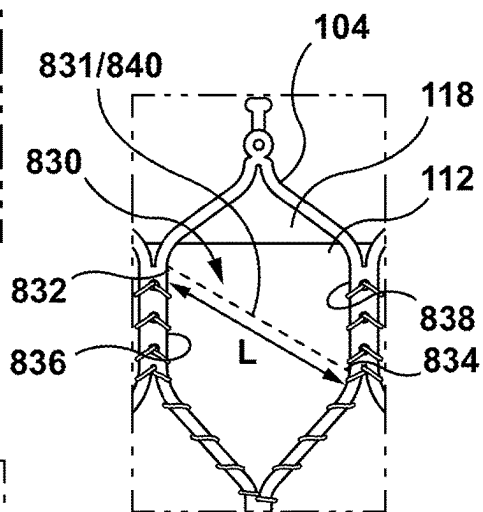
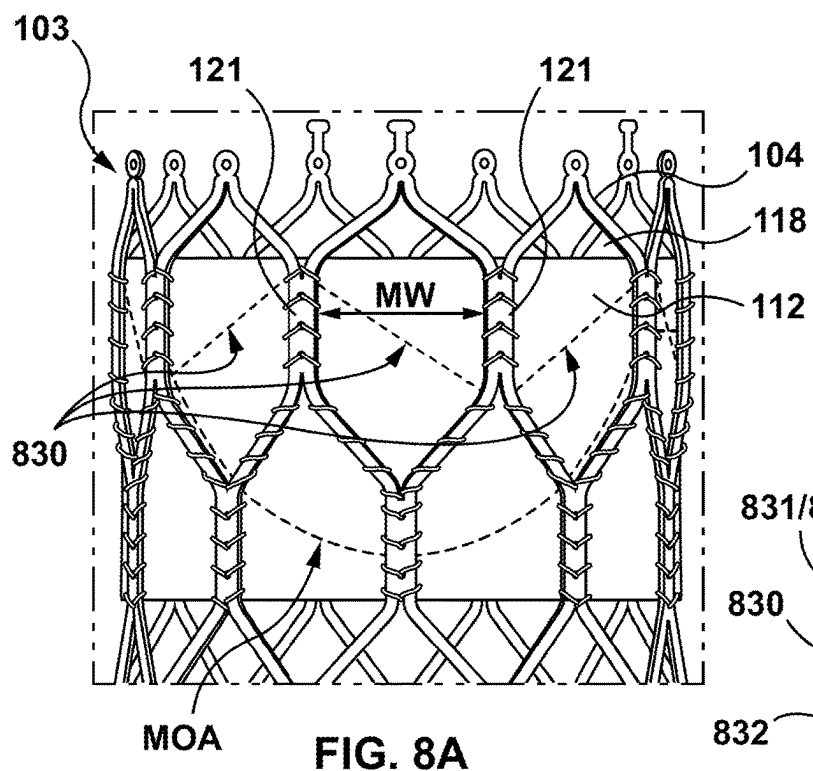
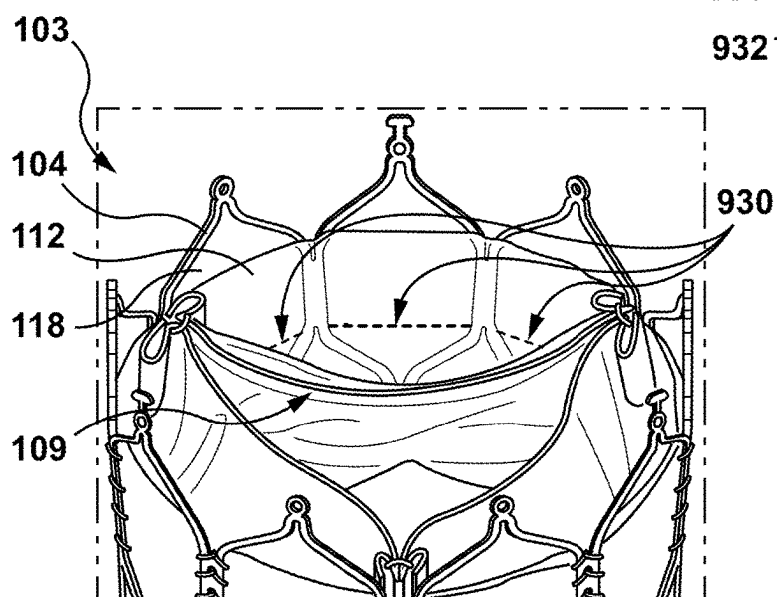
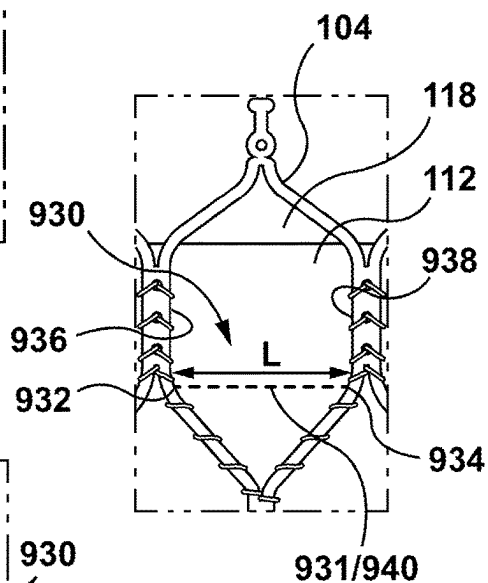
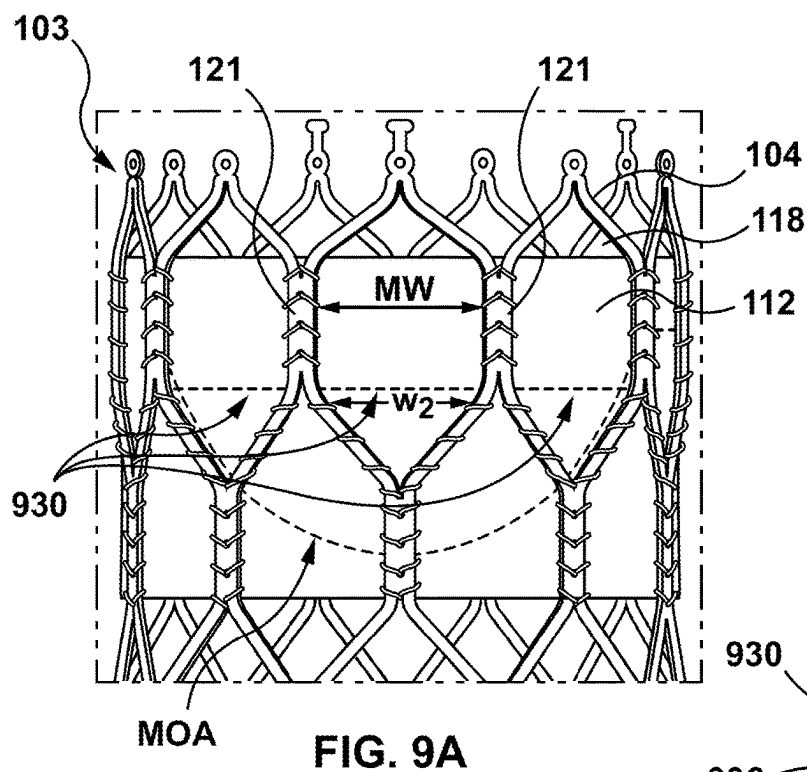
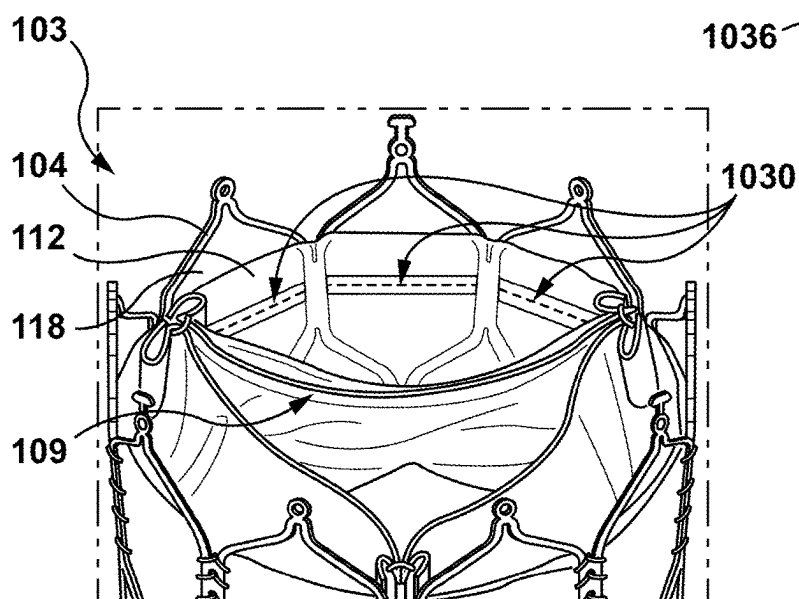
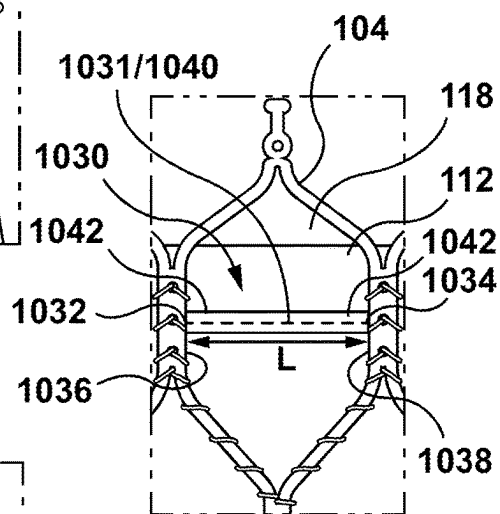
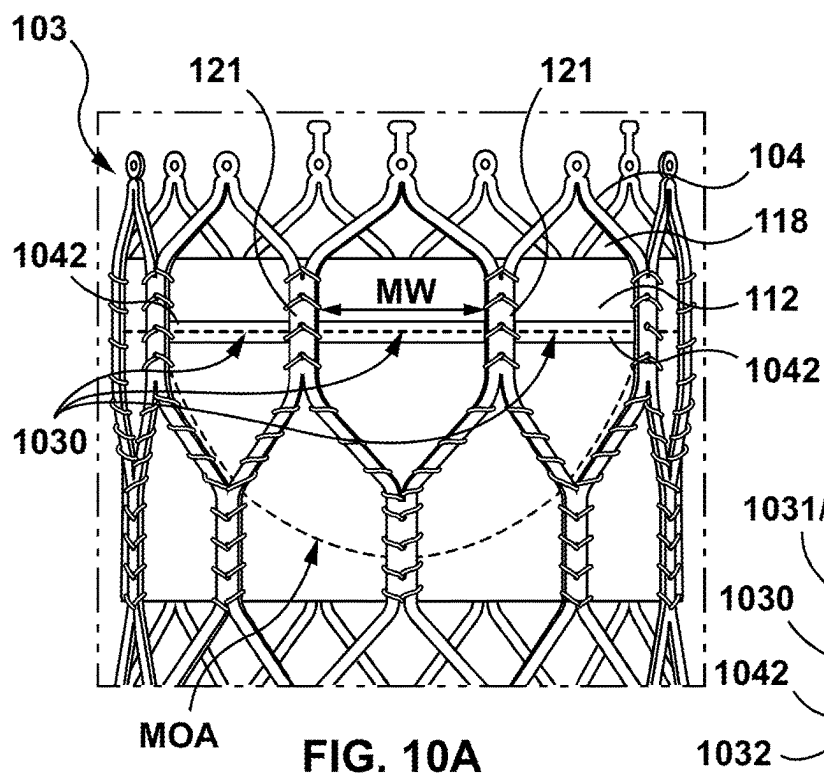


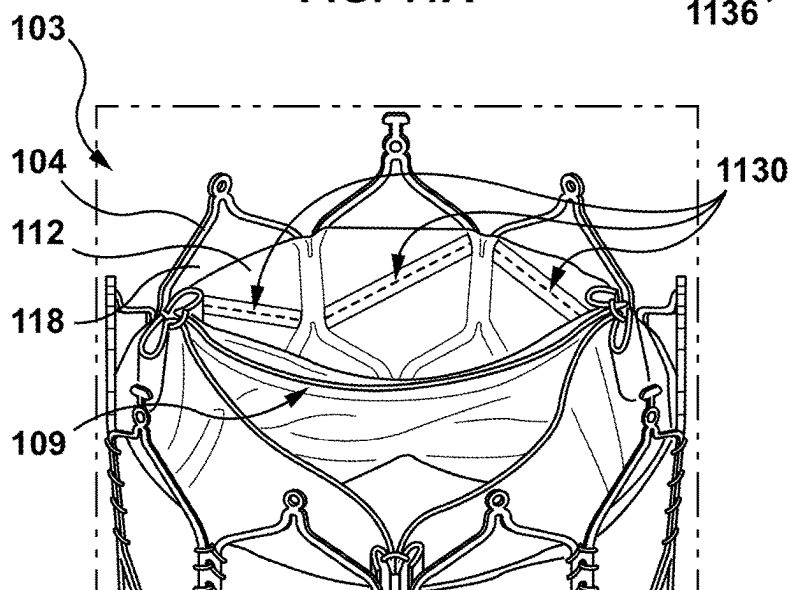
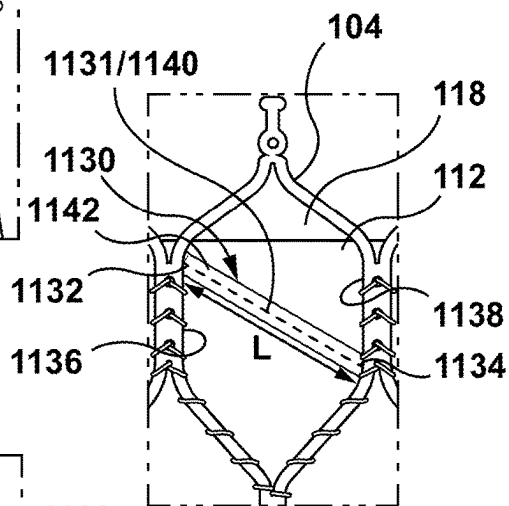
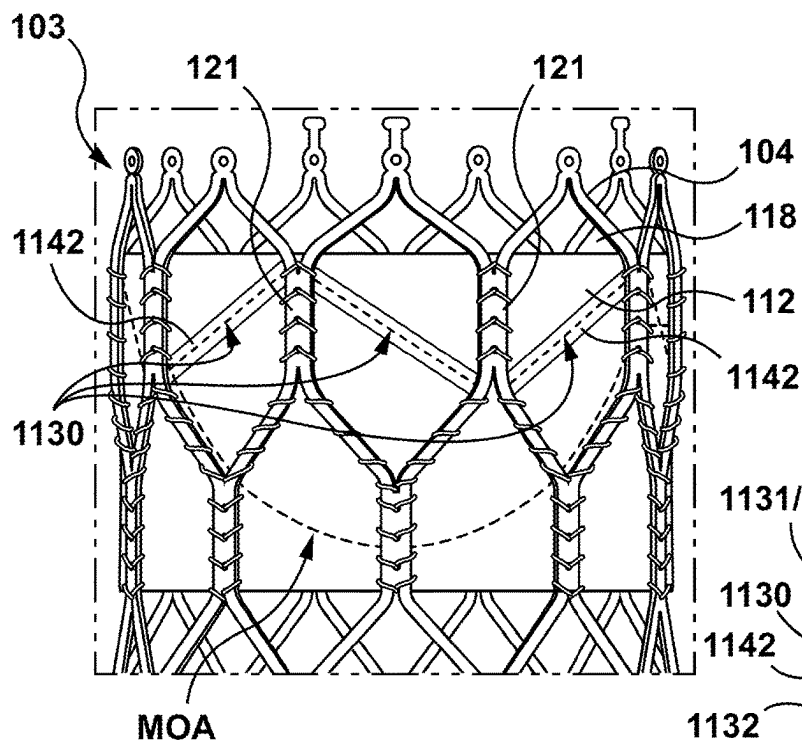
FIG. 6B











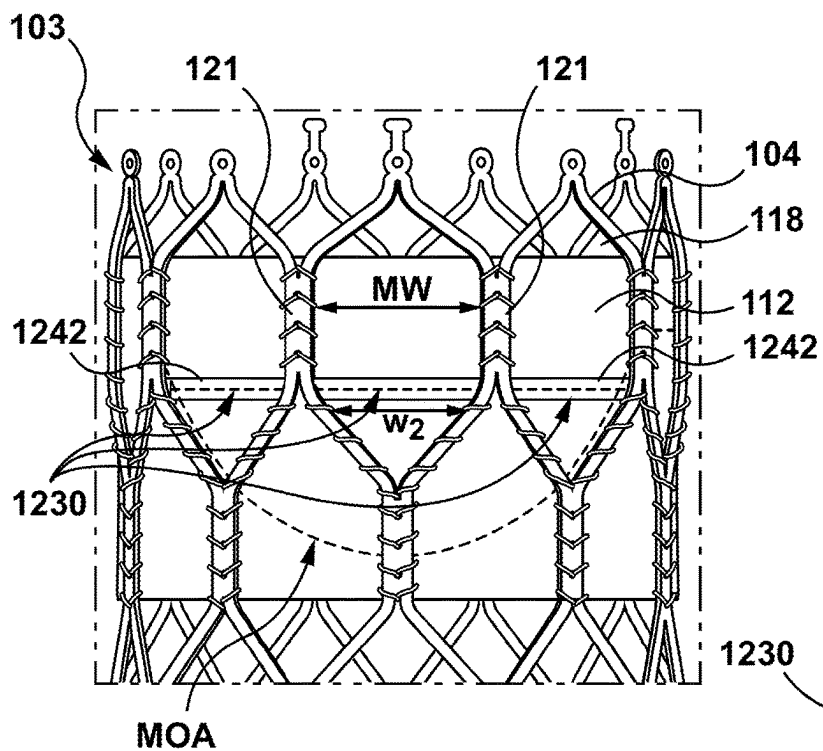


FIG. 12A

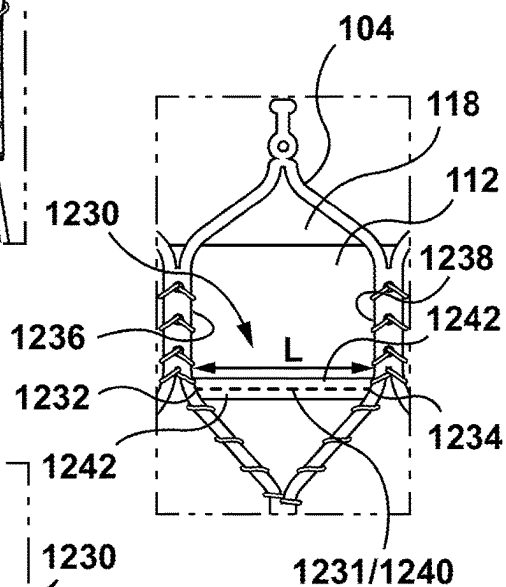


FIG. 12C

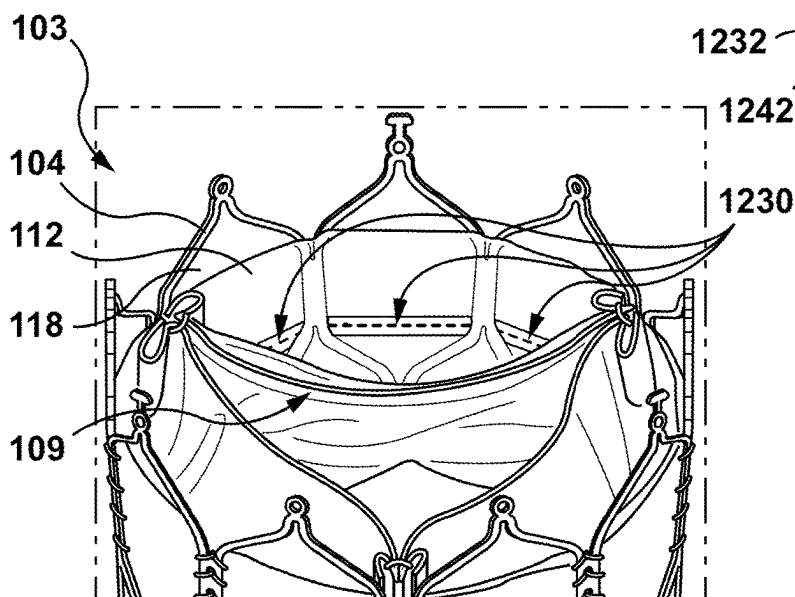


FIG. 12B

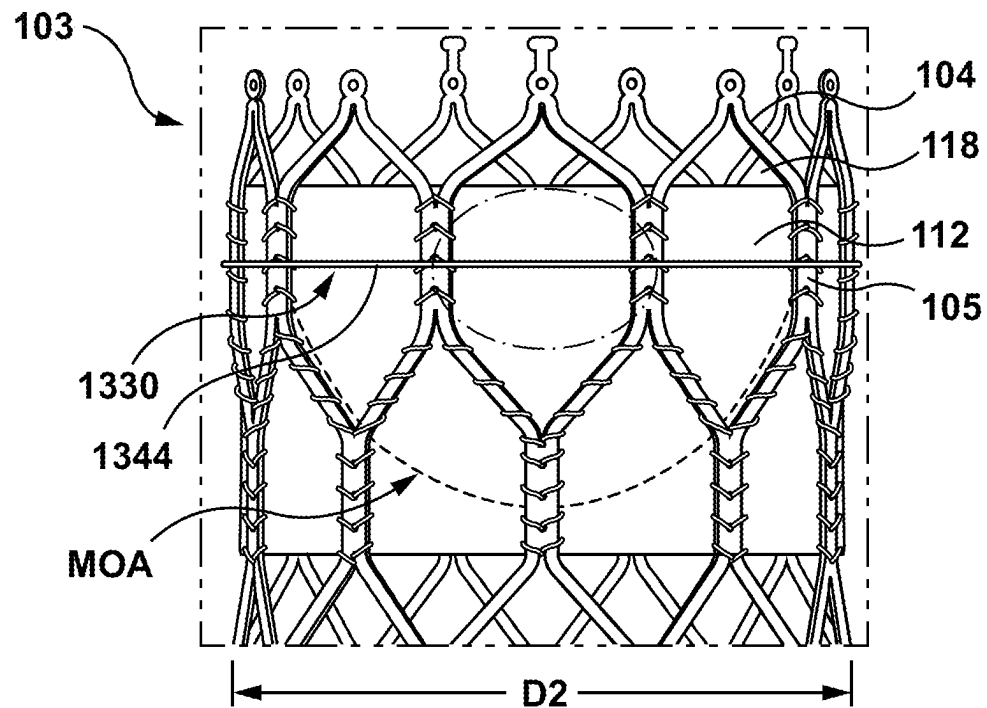


FIG. 13A

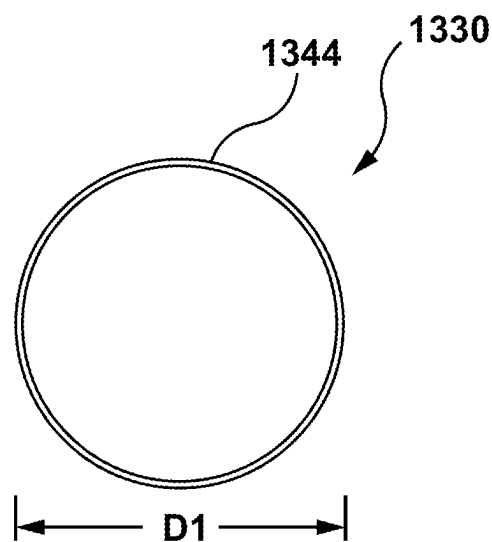


FIG. 13B

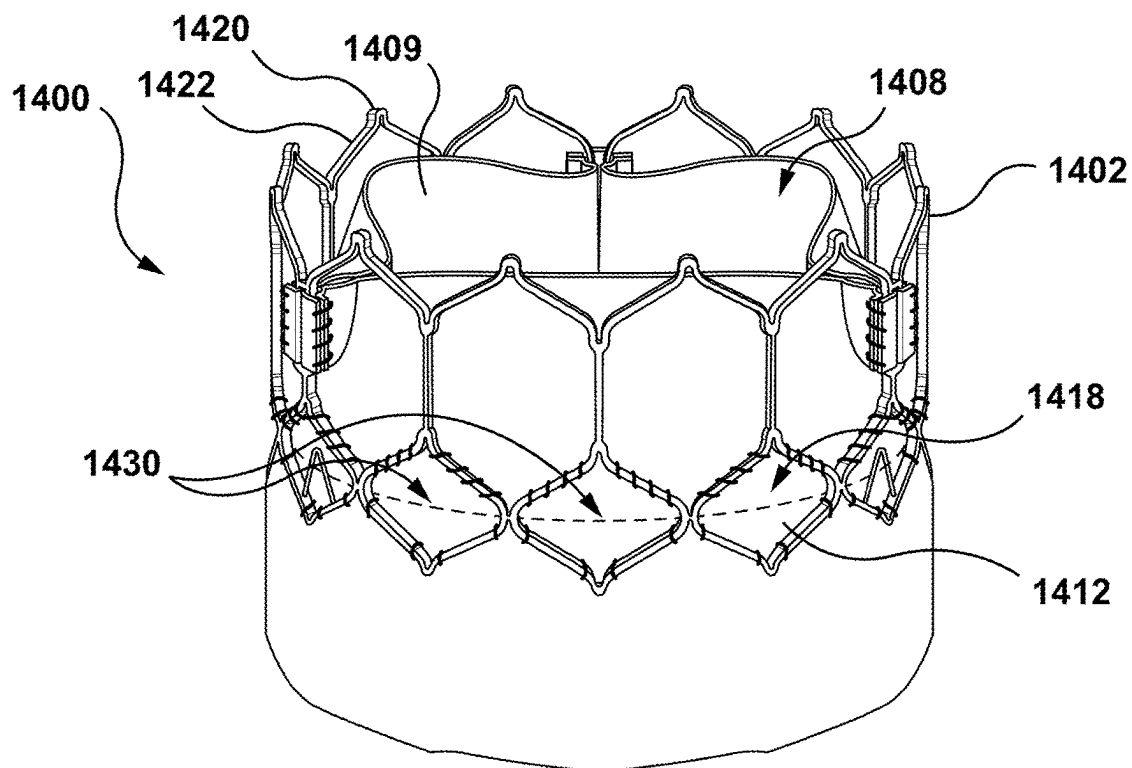


FIG. 14

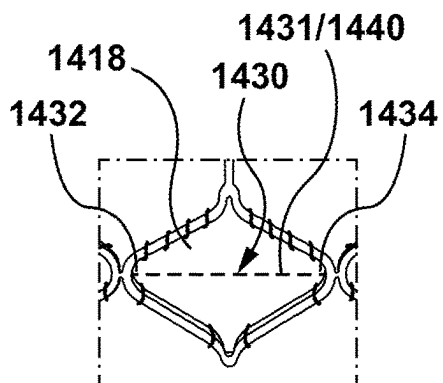


FIG. 14A

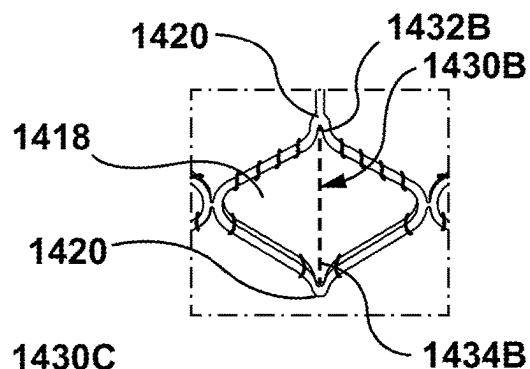


FIG. 14B

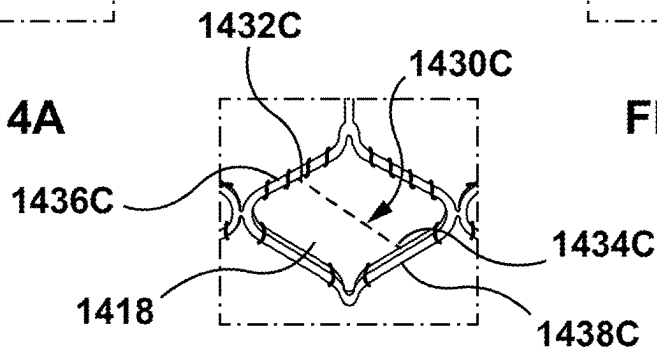


FIG. 14C

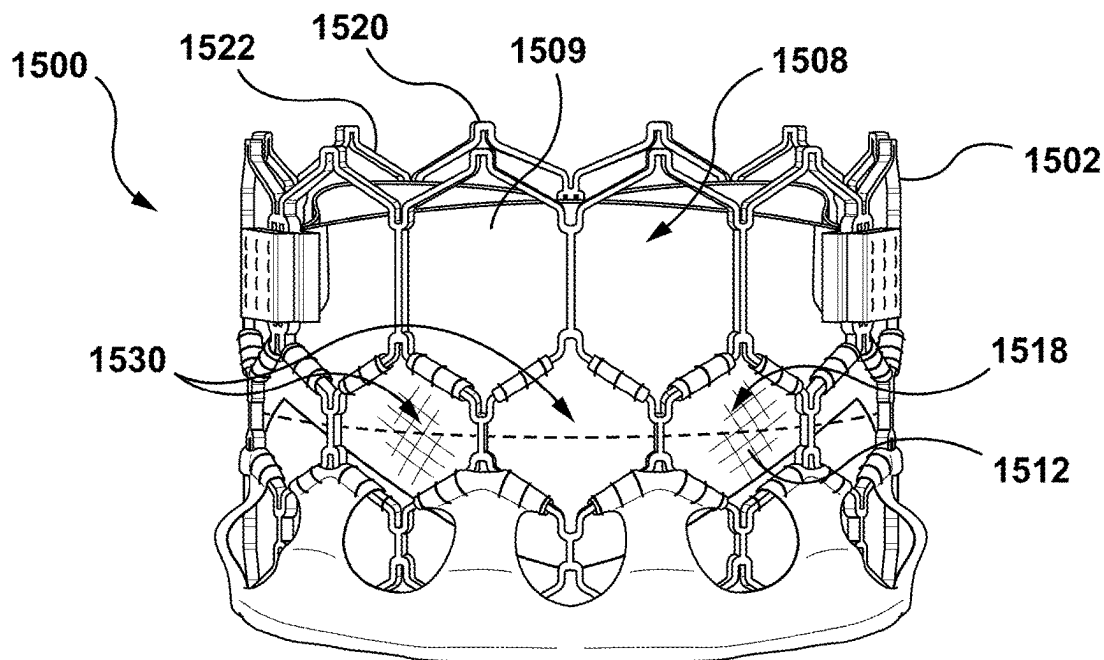


FIG. 15

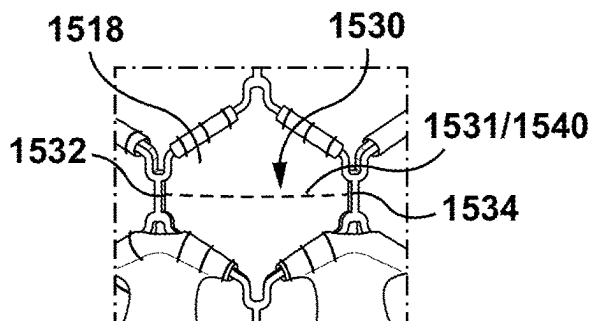


FIG. 15A

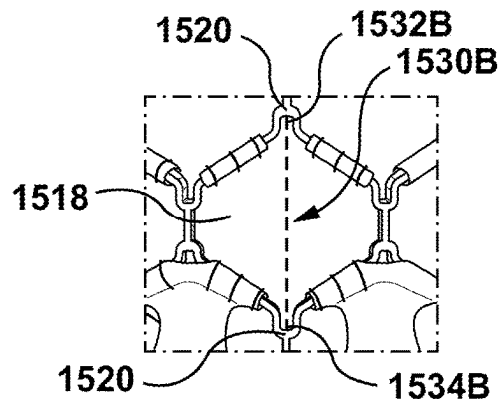


FIG. 15B

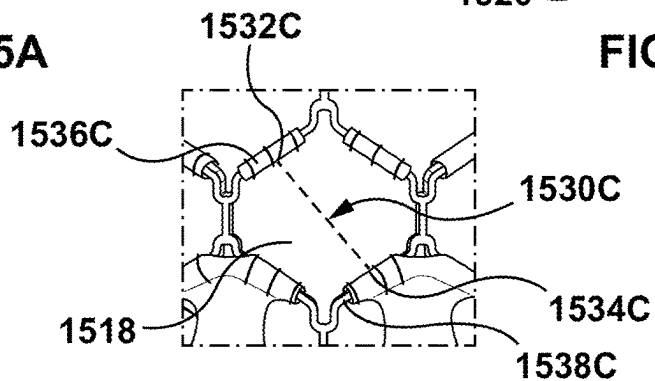


FIG. 15C

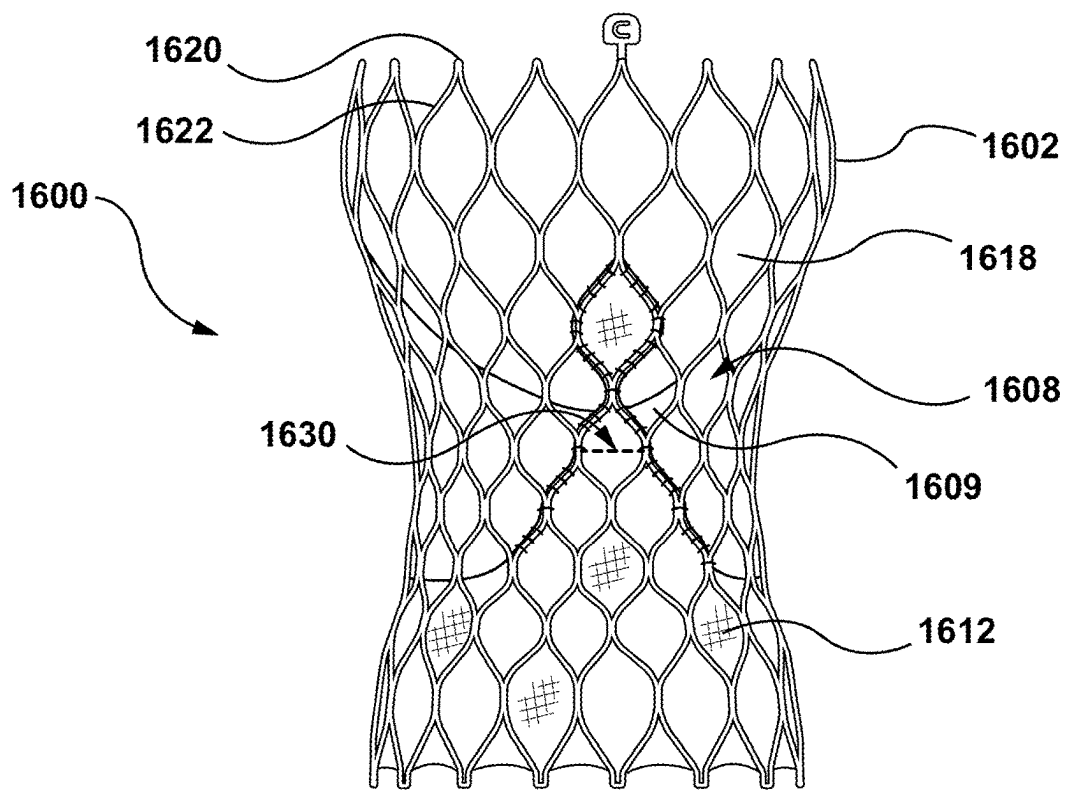


FIG. 16

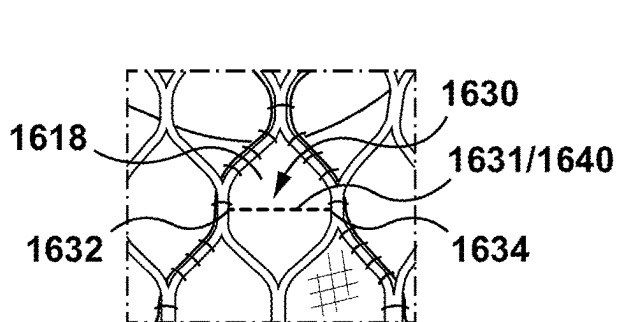


FIG. 16A

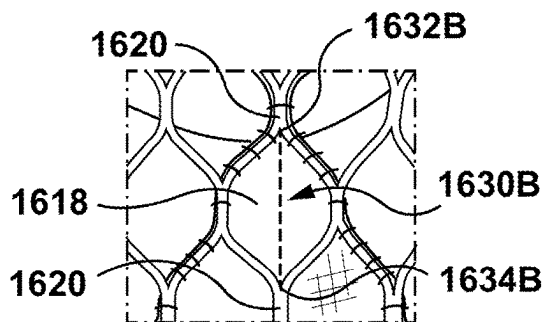


FIG. 16B

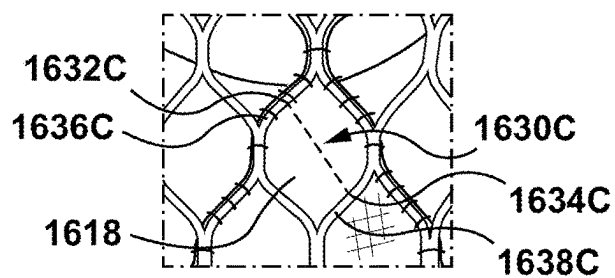


FIG. 16C

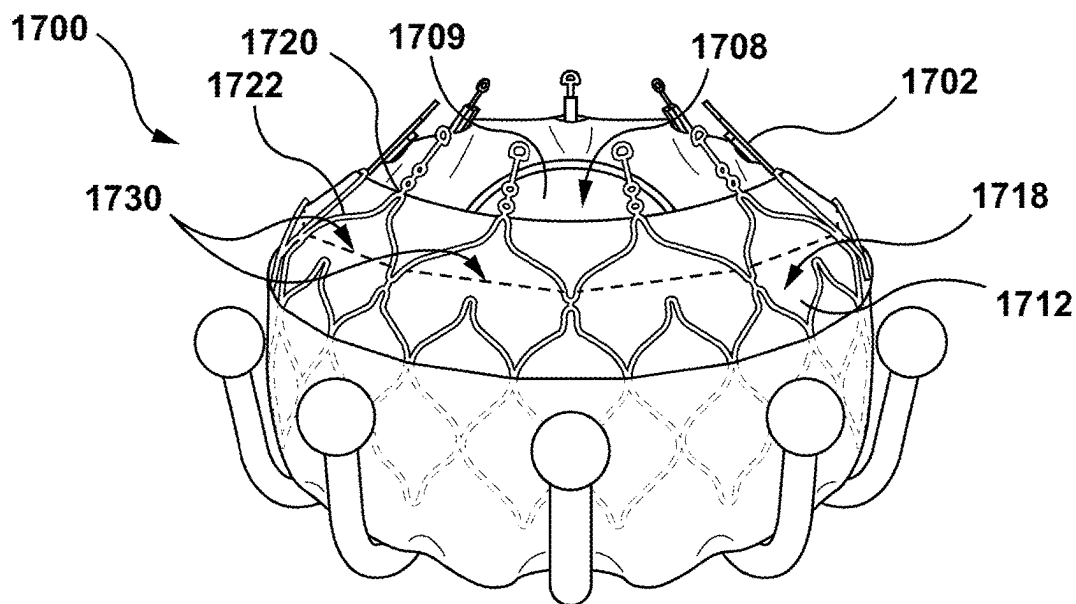


FIG. 17

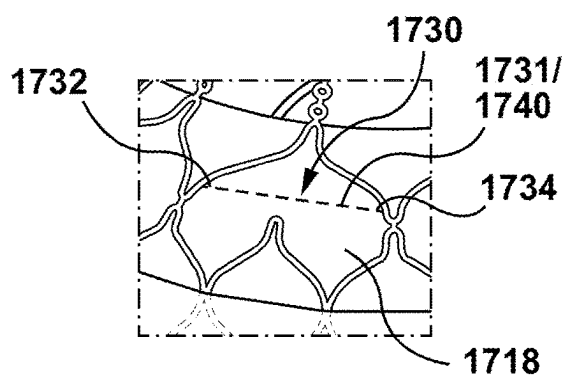


FIG. 17A

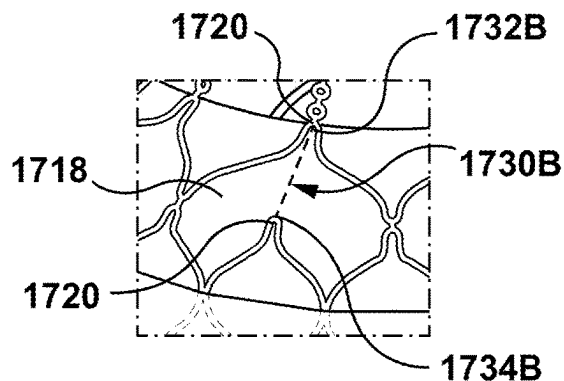


FIG. 17B

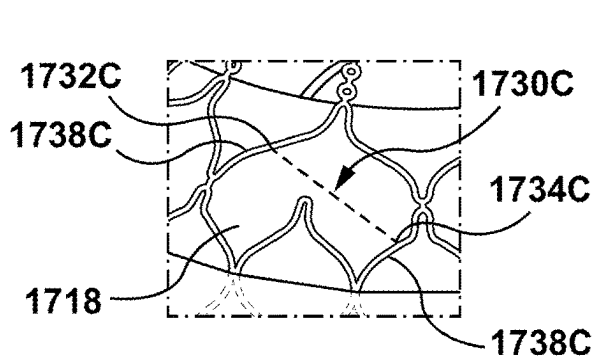


FIG. 17C

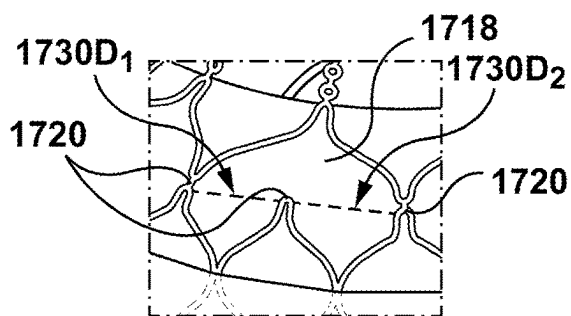


FIG. 17D

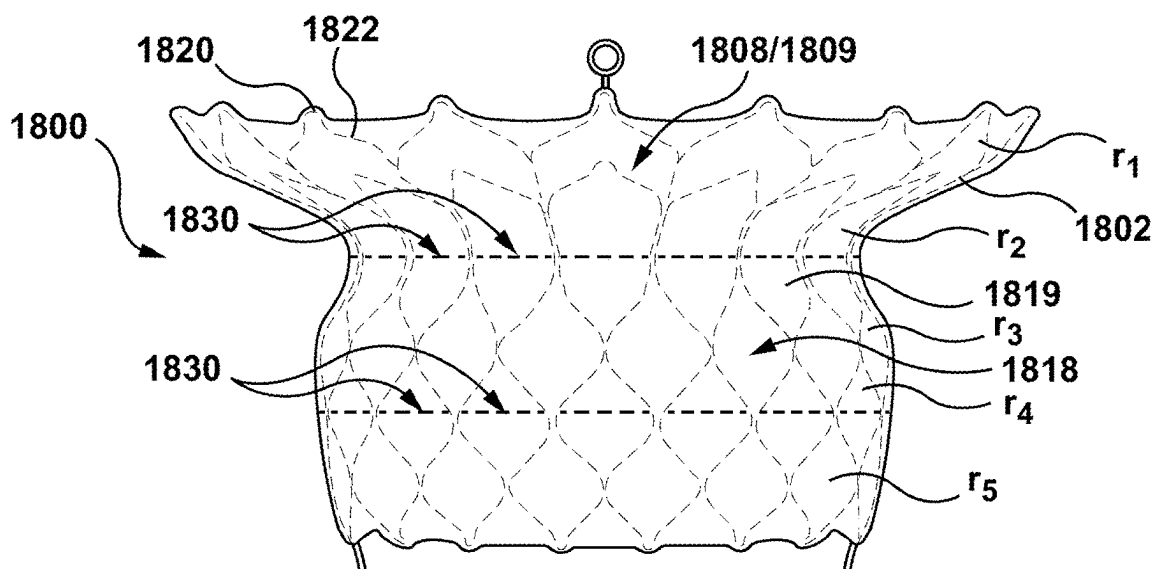


FIG. 18

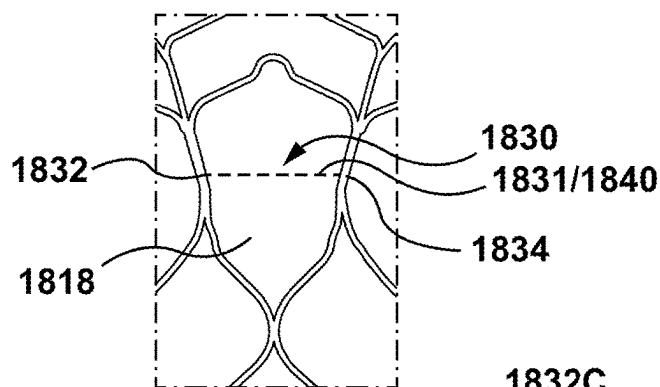


FIG. 18A

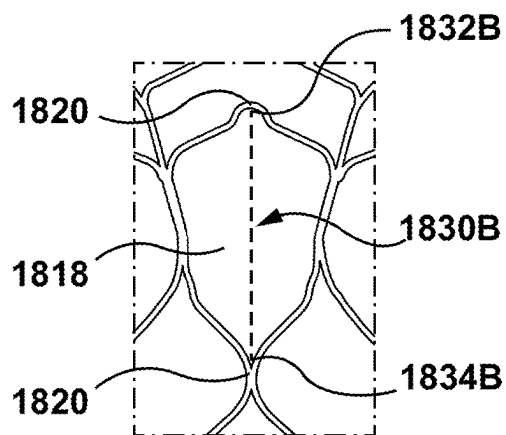


FIG. 18B

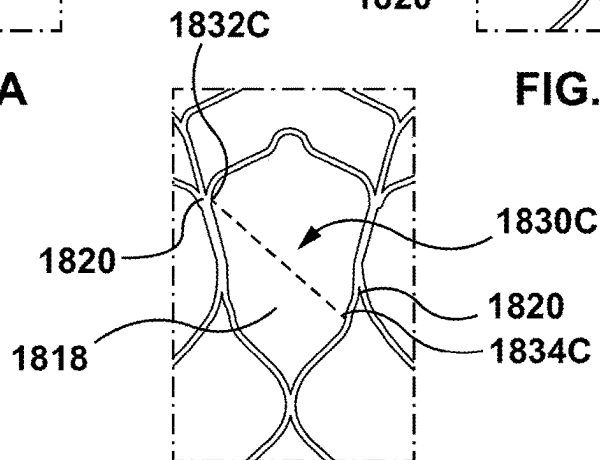


FIG. 18C

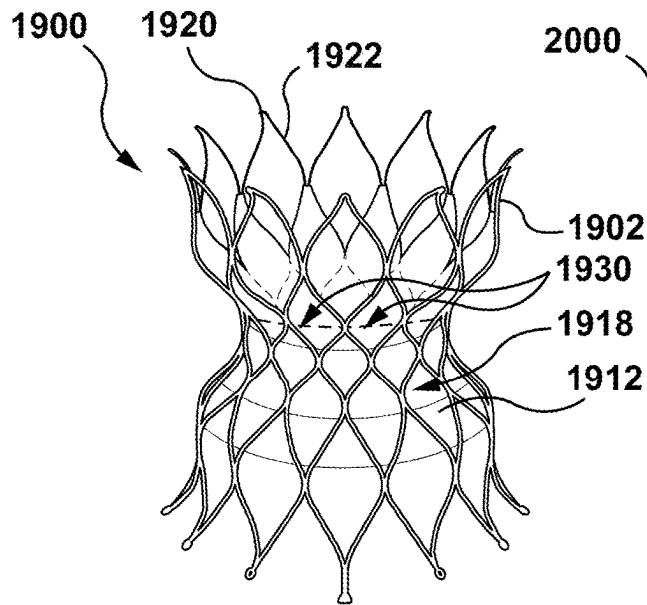


FIG. 19

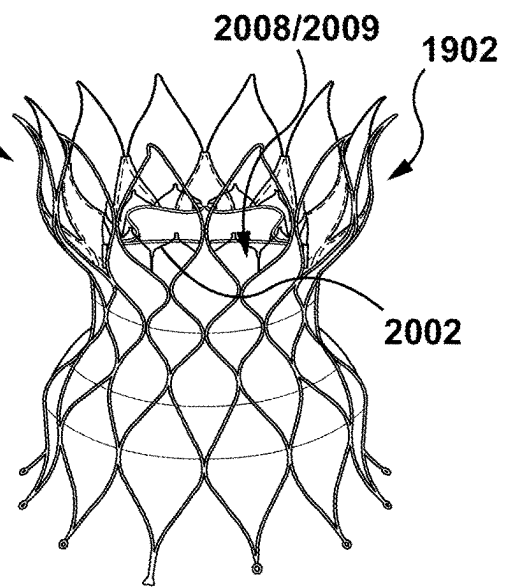


FIG. 20

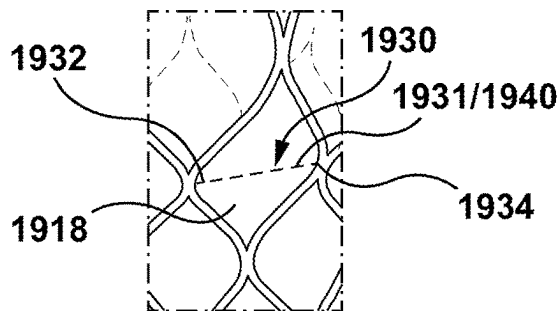


FIG. 19A

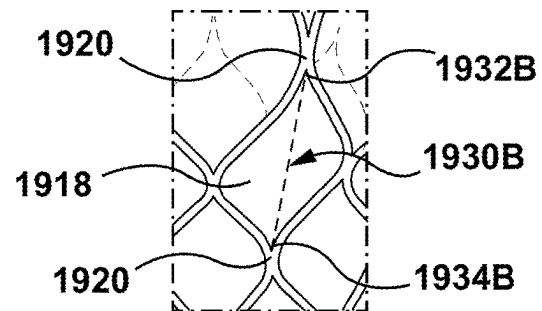


FIG. 19B

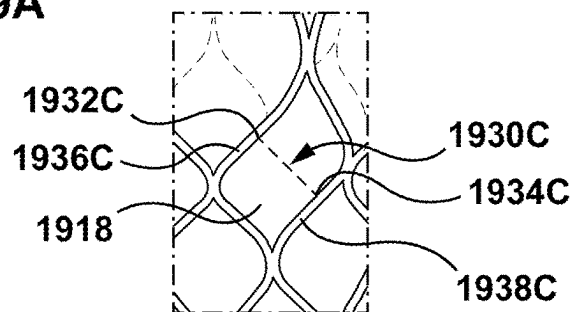


FIG. 19C

1

**SKIRT-REINFORCEMENT MEMBERS FOR
PROSTHETIC VALVE DEVICES****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/129,166, filed Dec. 22, 2020, which is hereby incorporated by reference in its entirety for all purposes.

FIELD

The present technology is generally related to prosthetic valve devices, and in particular is directed to prosthetic valve devices including a skirt.

BACKGROUND

The human heart is a four chambered, muscular organ that provides blood circulation through the body during a cardiac cycle. The four main chambers include the right atrium and right ventricle which supplies the pulmonary circulation, and the left atrium and left ventricle which supplies oxygenated blood received from the lungs into systemic circulation. To ensure that blood flows in one direction through the heart, atrioventricular valves (tricuspid and mitral valves) are present between the junctions of the atrium and the ventricles, and semi-lunar valves (pulmonary valve and aortic valve) govern the exits of the ventricles leading to the lungs and the rest of the body. These valves contain leaflets or cusps that open and shut in response to blood pressure changes caused by the contraction and relaxation of the heart chambers. The valve leaflets move apart from each other to open and allow blood to flow downstream of the valve, and coapt to close and prevent backflow or regurgitation in an upstream manner.

Diseases associated with heart valves, such as those caused by damage or a defect, can include stenosis and valvular insufficiency or regurgitation. For example, valvular stenosis causes the valve to become narrowed and hardened which can prevent blood flow to a downstream heart chamber from occurring at the proper flow rate and may cause the heart to work harder to pump the blood through the diseased valve. Valvular insufficiency or regurgitation occurs when the valve does not close completely, allowing blood to flow backwards, thereby causing the heart to be less efficient. A diseased or damaged valve, which can be congenital, age-related, drug-induced, or in some instances, caused by infection, can result in an enlarged, thickened heart that loses elasticity and efficiency. Some symptoms of heart valve diseases can include weakness, shortness of breath, dizziness, fainting, palpitations, anemia and edema, and blood clots which can increase the likelihood of stroke or pulmonary embolism. Symptoms can often be severe enough to be debilitating and/or life threatening.

Heart valve prostheses have been developed for repair and replacement of diseased and/or damaged heart valves. Such heart valve prostheses can be percutaneously delivered and deployed at the site of the diseased heart valve through catheter-based delivery systems. Such heart valve prostheses are delivered in a radially compressed or crimped configuration so that the heart valve prosthesis can be advanced through the patient's vasculature. Once positioned at the treatment site, the heart valve prosthesis is expanded to engage tissue at the diseased heart valve region to, for instance, hold the heart valve prosthesis in position.

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The present disclosure relates to improvements in a heart valve prosthesis to ensure that the heart valve prosthesis has a low profile for transcatheter delivery through a patient's vasculature.

SUMMARY

According to a first embodiment hereof, the present disclosure provides a prosthesis having a radially expanded configuration and a radially compressed configuration. The prosthesis includes a frame including a plurality of crowns and a plurality of struts with each crown being formed between a pair of opposing struts. A plurality of side openings is defined by edges of the plurality of crowns and the plurality of struts. The prosthesis also includes a skirt coupled to a surface of the frame. The skirt extends over at least one side opening of the plurality of side openings of the frame. The prosthesis also includes a skirt-reinforcement member that spans the at least one side opening from a first crown or strut edge defining the at least one side opening to a second crown or strut edge defining the at least one side opening. The skirt-reinforcement member has a first end, a second end, and a length therebetween. The first end of the skirt-reinforcement member is attached to the first crown or strut edge and the second end of the skirt-reinforcement member is attached to the second crown or strut edge. The skirt-reinforcement member is attached to the skirt along an unsupported portion of the skirt that spans the at least one side opening.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the skirt-reinforcement member includes a suture forming a plurality of stitches.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the skirt-reinforcement member includes tissue.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the skirt-reinforcement member includes fabric.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the at least one side opening is substantially diamond-shaped. In an embodiment, each side opening of the plurality of side openings is substantially diamond-shaped. In an embodiment, an end of the frame includes a row of side openings around a circumference of the frame, the row including between six and nine side openings.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the first end of the skirt-reinforcement member is offset from the second end of the skirt-reinforcement member such that the skirt-reinforcement member is angled relative to an axis of the frame.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the first end of the skirt-reinforcement member is aligned with the second end of the skirt-reinforcement member such that the skirt-reinforcement member extends substantially perpendicular relative to an axis of the frame. In an embodiment, the at least one side opening includes a maximum width and the skirt-reinforcement member spans across the at least one side opening at the maximum width. In an embodiment, the frame includes a plurality of nodes, each node being a region where two of the plurality of crowns of the frame meet, and the skirt-reinforcement member spans across the at least one side opening from one node to another node.

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In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the frame includes a row of side openings around a circumference of the frame and a skirt-reinforcement member spans each side opening of the row of side openings around the circumference of the frame.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the frame is an inner frame and the prosthesis further comprises an outer frame coupled to the inner frame, the outer frame having a greater diameter than the inner frame.

In an aspect of the first embodiment, and in combination with any other aspects herein, the disclosure provides that the prosthesis is a heart-valve prosthesis and the prosthesis further comprises a prosthetic valve component disposed within and secured to the frame, the prosthetic valve being configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame. In an embodiment, the heart-valve prosthesis is configured for placement within a mitral valve in situ.

According to a second embodiment hereof, the present disclosure provides a prosthesis having a radially expanded configuration and a radially compressed configuration. The prosthesis includes a frame including a plurality of crowns and a plurality of struts with each crown being formed between a pair of opposing struts. A plurality of side openings is defined by edges of the plurality of crowns and the plurality of struts. The prosthesis also includes a skirt coupled to a surface of the frame. The skirt extends over at least one side opening of the plurality of side openings of the frame. The prosthesis also includes a skirt-reinforcement member that circumscribes an outer surface of the frame. A diameter of the skirt-reinforcement member is approximately the same as a diameter of the frame when the prosthesis is in the radially expanded configuration. The skirt-reinforcement member is attached to the skirt along an unsupported portion of the skirt that spans the at least one side opening of the plurality of side openings of the frame and the skirt-reinforcement member is not directly attached to the frame.

In an aspect of the second embodiment, and in combination with any other aspects herein, the disclosure provides that the skirt-reinforcement member is a ring of an elastic material.

In an aspect of the second embodiment, and in combination with any other aspects herein, the disclosure provides that the at least one side opening of the plurality of side openings is substantially diamond-shaped and is defined by at least two struts. In an embodiment, each side opening of the plurality of side openings is substantially diamond-shaped and is defined by at least four struts. In an embodiment, an end of the frame includes a row of side openings around a circumference of the frame, the row including between six and nine side openings.

In an aspect of the second embodiment, and in combination with any other aspects herein, the disclosure provides that the frame is an inner frame and the prosthesis further comprises an outer frame coupled to the inner frame, the outer frame having a greater diameter than the inner frame.

In an aspect of the second embodiment, and in combination with any other aspects herein, the disclosure provides that the prosthesis is a heart-valve prosthesis and the prosthesis further comprises a prosthetic valve component disposed within and secured to the frame, the prosthetic valve being configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame. In

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an embodiment, the heart-valve prosthesis is configured for placement within a mitral valve in situ.

The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

The foregoing and other features and advantages of the invention will be apparent from the following description of embodiments thereof as illustrated in the accompanying drawings. The accompanying drawings, which are incorporated herein and form a part of the specification, further serve to explain the principles of the invention and to enable a person skilled in the pertinent art to make and use the invention. The drawings are not to scale.

FIG. 1 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with an aspect of the disclosure.

FIG. 2 depicts a perspective view of a valve support of the transcatheter heart valve prosthesis of FIG. 1 with a prosthetic valve component secured therein in accordance with an aspect of the disclosure.

FIG. 3 depicts an atrial end view of the transcatheter heart valve prosthesis shown in FIG. 1 in accordance with an aspect of the disclosure.

FIG. 4 depicts a ventricular end view of the transcatheter heart valve prosthesis shown in FIG. 1 in accordance with an aspect of the disclosure.

FIG. 5 is an enlarged side view of a side opening of the valve support of FIG. 2.

FIG. 6A is a perspective atrial end view of the transcatheter heart valve prosthesis shown in FIG. 1 in accordance with an aspect of the disclosure.

FIG. 6B is an atrial end view of the transcatheter heart valve prosthesis shown in FIG. 1 in accordance with an aspect of the disclosure.

FIG. 7A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with an aspect of the disclosure.

FIG. 7B is a perspective atrial end view of the valve support of FIG. 7A.

FIG. 7C is an enlarged view of a side opening of the valve support of FIG. 7A.

FIG. 8A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with another aspect of the disclosure.

FIG. 8B is a perspective atrial end view of the valve support of FIG. 8A.

FIG. 8C is an enlarged view of a side opening of the valve support of FIG. 8A.

FIG. 9A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with another aspect of the disclosure.

FIG. 9B is a perspective atrial end view of the valve support of FIG. 9A.

FIG. 9C is an enlarged view of a side opening of the valve support of FIG. 9A.

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FIG. 10A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with another aspect of the disclosure.

FIG. 10B is a perspective atrial end view of the valve support of FIG. 10A.

FIG. 10C is an enlarged view of a side opening of the valve support of FIG. 10A.

FIG. 11A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with another aspect of the disclosure.

FIG. 11B is a perspective atrial end view of the valve support of FIG. 11A.

FIG. 11C is an enlarged view of a side opening of the valve support of FIG. 11A.

FIG. 12A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support further includes a plurality of skirt-reinforcement members in accordance with another aspect of the disclosure.

FIG. 12B is a perspective atrial end view of the valve support of FIG. 12A.

FIG. 12C is an enlarged view of a side opening of the valve support of FIG. 12A.

FIG. 13A is a side view of the valve support of the transcatheter heart valve prosthesis of FIG. 1, wherein the valve support includes a skirt-reinforcement member that circumscribes an outer surface of the valve support.

FIG. 13B is a top view of the skirt-reinforcement member of FIG. 13A, wherein the skirt-reinforcement member is removed from the valve support for purposes of illustration only.

FIG. 14 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with another aspect of the disclosure, wherein the frame further includes a plurality of skirt-reinforcement members.

FIG. 14A is an enlarged view of a side opening of the transcatheter heart valve prosthesis of FIG. 14.

FIG. 14B is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 14C is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 15 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with another aspect of the disclosure, wherein the frame further includes a plurality of skirt-reinforcement members.

FIG. 15A is an enlarged view of a side opening of the transcatheter heart valve prosthesis of FIG. 15.

FIG. 15B is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 15C is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 16 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with another aspect of the disclosure, wherein the frame further includes a plurality of skirt-reinforcement members.

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FIG. 16A is an enlarged view of a side opening of the transcatheter heart valve prosthesis of FIG. 16.

FIG. 16B is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 16C is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 17 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with another aspect of the disclosure, wherein the frame further includes a plurality of skirt-reinforcement members.

FIG. 17A is an enlarged view of a side opening of the transcatheter heart valve prosthesis of FIG. 17.

FIG. 17B is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 17C is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 17D is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein two skirt-reinforcement members are disposed over a single side opening of the transcatheter valve prosthesis.

FIG. 18 depicts a perspective view of a transcatheter heart valve prosthesis in accordance with another aspect of the disclosure, wherein the frame further includes a plurality of skirt-reinforcement members.

FIG. 18A is an enlarged view of a side opening of the transcatheter heart valve prosthesis of FIG. 18.

FIG. 18B is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 18C is an enlarged view of a side opening of a transcatheter heart valve prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 19 depicts a perspective view of a docking prosthesis in accordance with another aspect of the disclosure, wherein a frame of the docking prosthesis includes a plurality of skirt-reinforcement members.

FIG. 19A is an enlarged view of a side opening of the docking prosthesis of FIG. 19.

FIG. 19B is an enlarged view of a side opening of a docking prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in a vertical orientation.

FIG. 19C is an enlarged view of a side opening of a docking prosthesis according to another embodiment hereof, wherein a skirt-reinforcement member extends in an angled orientation.

FIG. 20 is a perspective view of the docking prosthesis of FIG. 19 and a valve prosthesis disposed in the docking prosthesis.

DETAILED DESCRIPTION

Specific embodiments of the present invention are now described with reference to the figures, wherein like reference numbers indicate identical or functionally similar elements. The terms “distal” and “proximal”, when used in the

following description to refer to a native vessel, native valve, or a device to be implanted into a native vessel or native valve, such as a transcatheter heart valve prosthesis, are with reference to the direction of blood flow. Thus, “distal” and “distally” refer to positions in a downstream direction with respect to the direction of blood flow and the terms “proximal” and “proximally” refer to positions in an upstream direction with respect to the direction of blood flow.

Embodiments hereof relate to a skirt-reinforcement member for supporting or reinforcement a skirt that spans across a side opening of a stent or frame of a valve prosthesis. The skirt-reinforcement member provides the valve prosthesis with a reinforced composite skirt having long-lasting durability and superior implant performance. In addition, as will be explained in more detail herein, the skirt-reinforcement member is configured to prevent billowing of the skirt material that spans across the side opening of the frame of the valve prosthesis, as such billowing may undesirably result in contact between the skirt and the leaflets of the valve prosthesis after the valve prosthesis is deployed in situ. If the leaflets of the valve prosthesis contact the skirt during opening and closing in situ, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion due to the undesired billowing of the skirt. Additionally, the greater relative motion between the skirt and the frame may further induce early skirt abrasion. Early leaflet tissue abrasion and/or early skirt abrasion has a negative impact on the long-term durability of the valve prosthesis. The skirt-reinforcement members disclosed herein reinforce the material of the skirt that spans across the side opening of the frame of the valve prosthesis to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt and the leaflets.

FIGS. 1-6 illustrate a transcatheter heart valve prosthesis **100** that may be utilized with embodiments of skirt-reinforcement members described herein. The heart valve prosthesis **100** is illustrated herein in order to facilitate description of the present invention. The following description of the transcatheter heart valve prosthesis **100** is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. It is understood that any number of alternate heart valve prostheses can be used with the skirt-reinforcement members described herein. Other non-limiting examples of transcatheter heart valve prostheses that can be used with the skirt-reinforcement members described herein are described in U.S. application Ser. No. 16/853,851 to McVeigh et al., U.S. Pat. No. 9,034,032 to McLean et al. and International Patent Application No. PCT/US2014/029549 to McLean et al., U.S. Patent Application Publication No. 2012/0101572 to Kovalsky et al., U.S. Patent Application Publication No. 2012/0035722 to Tuval, U.S. Patent Application Publication No. 2006/0265056 to Nguyen et al., U.S. Patent Application Publication No. 2007/05409266 to Birdsall, and U.S. Patent Application Publication No. 2007/05409269 to Dolan et al., each of which is incorporated by reference herein in its entirety. Although the transcatheter heart valve prosthesis **100** is a heart valve prosthesis configured for placement within a mitral heart valve, embodiments of skirt-reinforcement members described herein may be utilized with any valve prosthesis having a skirt. For example, embodiments of skirt-reinforcement members described herein may be utilized with a transcatheter heart valve configured for placement within a pulmonary, aortic, mitral, or tricuspid valve, or may be utilized with a transcatheter valve prosthesis configured for placement within a venous valve or

within other body passageways where it is deemed useful. There is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. In addition, embodiments of skirt-reinforcement members described herein may be utilized with any stent or frame having a skirt for which reinforcement thereof is desirable to limit billowing, and it is not required that the stent or frame include a prosthetic valve component disposed therein.

A perspective view of the transcatheter heart valve prosthesis **100** in accordance with an aspect of the disclosure is shown in FIG. 1. The transcatheter heart valve prosthesis **100** is configured to be radially compressed into a reduced-diameter crimped configuration for delivery within a vasculature (not shown) and to return to an expanded, deployed configuration, as shown in FIG. 1. Stated another way, the transcatheter heart valve prosthesis **100** has a crimped configuration for delivery within a vasculature and an expanded configuration for deployment within a native heart valve. In accordance with embodiments hereof, when in the crimped configuration, the transcatheter heart valve prosthesis **100** has a low profile suitable for delivery to and deployment within a native heart valve via a suitable delivery catheter that may be tracked to the deployment site of the native heart valve of a heart via any one of a transseptal, retrograde, or transapical approach. The transcatheter heart valve prosthesis **100** includes a stent or frame **102** and a prosthetic valve component **108** including at least one leaflet disposed within and secured to the frame **102**.

Any portion of the frame **102** described herein as an element of a heart valve prosthesis **100** may be made from any number of suitable biocompatible materials, e.g., stainless steel, nickel titanium alloys such as Nitinol™, cobalt chromium alloys such as MP35N, other alloys such as ELGILOY® (Elgin, Ill.), various polymers, pyrolytic carbon, silicone, polytetrafluoroethylene (PTFE), or any number of other materials or combination of materials. A suitable biocompatible material would be selected to provide the transcatheter heart valve prosthesis **100** to be configured to be compressed into a reduced-diameter crimped configuration for transcatheter delivery to a native valve, whereby release from a delivery catheter returns the prosthesis to an expanded, deployed configuration.

In an aspect of the disclosure, the frame **102** of the transcatheter heart valve prosthesis **100** includes a valve support **104** at least partially surrounded by and coupled to an anchor element **106**. The valve support **104** is a tubular stent-like or frame structure that defines a central lumen **110** from an inflow end **101** of the valve support **104** to an outflow end **103** of the valve support **104**. The valve support **104** is configured to support the prosthetic valve component **108** therein, which will be described in more detail below. In an embodiment, the valve support **104** has a substantially cylindrical shape in which the outflow end **103** of the valve support **104** has a diameter that is substantially the same as a diameter of the inflow end **101** of the valve support **104**.

The valve support **104** includes a skirt **112** coupled to a surface thereof. More particularly, the skirt **112** is coupled to an inner surface of the valve support **104** to line a portion thereof. Alternatively, the skirt **112** may be coupled to an outer surface of the valve support **104** to enclose a portion thereof as would be known to one of ordinary skill in the art of prosthetic valve construction. The skirt **112** may be a natural or biological material such as pericardium or another membranous tissue such as intestinal submucosa. Alternatively, the skirt **112** may be a low-porosity woven fabric,

such as polyester, Dacron fabric, or PTFE, which creates a one-way fluid passage when attached to the stent. In one embodiment, the skirt **112** may be a knit or woven polyester, such as a polyester or PTFE knit, which can be utilized when it is desired to provide a medium for tissue ingrowth and the ability for the fabric to stretch to conform to a curved surface. Polyester velour fabrics may alternatively be used, such as when it is desired to provide a medium for tissue ingrowth on one side and a smooth surface on the other side. These and other appropriate cardiovascular fabrics are commercially available from Bard Peripheral Vascular, Inc. of Tempe, Ariz., for example.

In an aspect of the disclosure, the anchor element **106** is a stent-like or frame structure that functions as an anchor for the transcatheter heart valve prosthesis **100** to secure its deployed position within a native annulus. The anchor element **106** is a substantially cylindrically-shaped structure that is configured to engage heart tissue at or below an annulus of a native heart valve, such as an annulus of a native mitral valve. At the inflow end **101** of the valve support **104**, the anchor element **106** is radially spaced a distance **S** from the valve support **104** to mechanically isolate the inflow end **101** of the valve support **104** from the anchor element **106**. The anchor element **106** includes one or more cleats or prongs **114** that extend outward from an exterior side thereof to engage heart tissue. In another embodiment, the anchor element **106** may employ barbs, spikes, or other tissue fixation mechanisms for engaging heart tissue.

The transcatheter heart valve prosthesis **100** further includes a brim or rim element **116** that extends outwardly from an upstream end of the anchor element **106**. The brim element **116** includes overlapping, 180 degree out of phase sinusoidal wire forms that are attached and hinged to the anchor element **106** by a suitable biocompatible low-profile fabric **117** used in bioprosthetic implants namely endovascular grafts, heart valves or left atrial appendage devices to promote bio-integration, such as woven polyethylene terephthalate (PET) fabric. The brim element **116** may act as an atrial retainer, if present, and to serve such a function the brim element **116** may be configured to engage tissue above a native annulus, such as a supra-annular surface or some other tissue in the left atrium, to thereby inhibit downstream migration of a prosthetic heart valve **100**, for e.g., during atrial systole.

The prosthetic valve component **108** of the transcatheter heart valve prosthesis **100** is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. FIGS. 1-6 illustrate an exemplary prosthetic valve component having three leaflets, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. When deployed in situ, the prosthetic valve component **108** in a closed state is configured to block blood flow in one direction to regulate blood flow through the central lumen **110** of the valve support **104**. FIG. 2 depicts a perspective view of the valve support **104** with a prosthetic valve component **108** secured therein, the valve support **104** being shown in FIG. 2 removed from the remainder of the transcatheter heart valve prosthesis **100** shown in FIG. 1 for ease of illustration. FIG. 3 depicts an atrial or inflow end view of the transcatheter heart valve prosthesis **100** shown in FIG. 1, and FIG. 4 depicts a ventricular or outflow end view of the transcatheter heart valve prosthesis **100** shown in FIG. 1. The prosthetic valve component **108** includes valve leaflets **109**, e.g., three valve leaflets **109**, that are disposed to coapt within an upstream portion of the valve support **104** with leaflet commissures

109A, **109B**, **109C** of the valve leaflets **109** being secured within a downstream portion of the valve support **104**, such that the valve leaflets **109** open during diastole. Leaflets **109** are attached along their bases to the valve support **104**, for example, using sutures or a suitable biocompatible adhesive. A margin of attachment, or MOA, is formed at the junction of the leaflets **109** to the valve support **104**. Adjoining pairs of leaflets **109** are attached to one another at their lateral ends to form leaflet commissures **109A**, **109B**, **109C**. The orientation of the leaflets **109** within the valve support **104** depends upon on which end of the transcatheter heart valve prosthesis **100** is the inflow end and which end of the transcatheter heart valve prosthesis **100** is the outflow end, thereby ensuring one-way flow of blood through the transcatheter heart valve prosthesis **100**.

The valve leaflets **109** may be attached to the skirt **112**. The valve leaflets **109** may be formed of various flexible materials including, but not limited to natural pericardial material such as tissue from bovine, equine or porcine origins, or synthetic materials such as polytetrafluoroethylene (PTFE), DACRON® polyester, pyrolytic carbon, or other biocompatible materials. With certain prosthetic leaflet materials, it may be desirable to coat one or both sides of the replacement valve leaflet with a material that will prevent or minimize overgrowth. It is further desirable that the prosthetic leaflet material is durable and not subject to stretching, deforming, or fatigue.

For delivery, the transcatheter heart valve prosthesis **100** is radially compressed into a reduced-diameter crimped configuration onto a delivery system for delivery within a vasculature. As known in the art, the delivery system includes an inner shaft that receives the transcatheter heart valve prosthesis **100** on a distal portion thereof and an outer sheath or capsule that is configured to compressively retain the transcatheter heart valve prosthesis **100** on the distal portion of the inner shaft during delivery. Stated another way, the outer sheath or capsule surrounds and constrains the transcatheter heart valve prosthesis **100** in the radially compressed or crimped configuration. An exemplary delivery system for delivering the transcatheter heart valve prosthesis **100** is described in U.S. Pat. No. 9,034,032 to McLean et al. and International Patent Application No. PCT/US2014/029549 to McLean et al, previously incorporated by reference herein. However, it will be apparent to one of ordinary skill in the art that other delivery systems may be utilized and that the components of the delivery system may vary depending upon the configuration and structure of the transcatheter valve prosthesis that is being delivered.

Referring to FIG. 2 as well as FIG. 5, the structure of the valve support **104** will now be described in more detail. FIG. 5 is an enlarged side view of a plurality of side openings **118** of the valve support **104**. The valve support **104** includes a plurality of crowns **120** and a plurality of struts **122** with each crown **120** being formed between a pair of opposing struts **122**. Each crown **120** is a curved segment or bend extending between opposing struts **122**. The valve support **104** is tubular, with the plurality of side openings **118** being defined by edges of the plurality of crowns **120** and the plurality of struts **122**. In an embodiment, the plurality of side openings **118** may be substantially diamond-shaped. The valve support **104** includes a plurality of nodes **121**. A node **121** is defined as a region where two crowns of the plurality of crowns **120** within the valve support **104** meet or connect. As best shown in FIG. 5, in this embodiment, the skirt **112** is attached to an inner surface of the valve support **104** around a circumference thereof. The skirt **112** lines the inner surface of the valve support **104** and spans across or

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extends over each side opening 118 of the plurality of side openings 118. Notably, as shown in FIG. 5, it is not required that the skirt 112 extend over the full opening of each side opening 118. Rather, the skirt 112 may span or cover only a portion of the side opening 118 as shown in FIG. 5.

A series of endmost inflow crowns 120A are formed at the inflow end 101 of the valve support 104, and a series of endmost outflow crowns 120B are formed at the outflow end 103 of the valve support 104. In an embodiment, the inflow end 101 of the valve support 104 has a total of nine endmost inflow crowns 120A around a circumference thereof. The inflow end 101 of the valve support 104 includes a row of side openings 118 around a circumference thereof, and the row has a total of nine side openings 118. Further, outflow end 103 of the valve support 104 has a total of nine endmost inflow crowns 120B around a circumference thereof. The outflow end 103 of the valve support 104 includes a row of side openings 118 around a circumference thereof, and the row has a total of nine side openings 118. In another embodiment hereof (not shown), each of the inflow end 101 and the outflow end 103 of the valve support 104 has between six and nine endmost inflow crowns 120A, 120B around a circumference thereof and includes the row of side openings 118 around a circumference thereof that includes between six and nine side openings 118.

A width W of the side openings 118 is relatively wider as compared to other stents or frames known in the art, thereby resulting a relatively lower total of side openings 118 around a circumference of the valve support 104. In an embodiment, width W is between $\frac{1}{24}^{th}$ and $\frac{1}{6}^{th}$ of the circumference of the valve support 104, or stated another way, between 4% and 16% of the circumference of the valve support 104. By increasing the width of the side openings 118, a lesser amount of material is required for the valve support 104 such that a lower profile may be achieved when the valve support 104 is crimped into a radially compressed configuration for delivery. More particularly, since the frame 102 includes both the valve support 104 and the anchor element 106, it is a challenge to reduce the profile of the transcatheter valve prosthesis 100 in the crimped or radially compressed configuration. The challenge with reducing the profile is that, in the crimped or radially compressed configuration, the incompressible material of the frame 102 imparts high compressive forces on the soft tissue material of the leaflets 109. Such high compressive forces may alter the integrity of the leaflets 109, thereby impacting the long-term durability of the transcatheter valve prosthesis 100. However, increasing the width W of the side openings 118 provides a reduction of the incompressible material of the frame 102, thereby enabling a lower profile in the crimped or radially compressed configuration.

However, reducing the incompressible material of the frame 102 means that the skirt 112 spans a longer distance between nodes 121 or between struts 122. Stated another way, with the width of the side openings 118 being relatively increased as described above, the amount of material of the skirt 112 that spans across the side openings 118 likewise increases and thus a greater amount of material of the skirt 112 is unattached to the valve support 104. Referring now to FIGS. 6A and 6B, when an increased amount of material of the skirt 112 spans across the side openings 118, there is an increased chance of the skirt 112 billowing or moving radially inwards towards the leaflets 109 as indicated by directional arrows 124. The skirt 112 may billow during valve opening and closing in situ, and the leaflets 109 may contact the skirt 112. Such billowing may undesirably result in contact between the skirt 112 and the leaflets 109 of the

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transcatheter heart valve prosthesis 100. If the leaflets 109 of the transcatheter heart valve prosthesis 100 contact the skirt 112 during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt 112 and the valve support 104 may further induce early skirt abrasion.

Embodiments hereof relate to skirt-reinforcement members that reinforce the material of the skirt 112 that spans across the side openings 118 of the valve support 104 to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt 112 and the leaflets 109. For sake of illustration, the skirt-reinforcement members described herein are incorporated onto the valve support 104, as the structure of the valve support 104 has already been described in detail above. However, as previously stated, the skirt-reinforcements members described herein may be incorporated onto any stent or frame having a skirt for which reinforcement thereof is desirable to limit billowing, and it is not required that the stent or frame include a prosthetic valve component disposed therein.

Turning to FIGS. 7A and 7B, a skirt-reinforcement member 1330 according to an aspect of the disclosure is illustrated. More particularly, FIG. 7A and FIG. 7B are side and perspective atrial end views, respectively, of the valve support 104 including a plurality of skirt-reinforcement members 730. In the embodiment of FIGS. 7A and 7B, the valve support 104 includes a row of side openings 118 around a circumference thereof and a skirt-reinforcement member 730 spans across or extends over each side opening 118 of the row of side openings 118 around the circumference of the valve support 104. However, it is not required that a skirt-reinforcement member 730 be utilized on every side opening 118 of the row of side openings 118 around the circumference of the valve support 104. Rather, a skirt-reinforcement member 730 may be utilized only on the side openings 118 having skirt material that may come into contact with leaflets 109 when the leaflets are opening and closing in situ. At least one side opening 118 must include a skirt-reinforcement member 730 according to an aspect of the present disclosure. Each skirt-reinforcement member 730 spans a side opening 118 from a first crown or strut edge defining the at least one side opening 118 to a second crown or strut edge defining the at least one side opening 118.

In the embodiment of FIGS. 7A and 7B, each skirt-reinforcement member 730 is a filament or strand 731 of suture that forms a plurality of stitches 740. Exemplary suture materials include but are not limited to a monofilament or plastic suture material, such as polypropylene. Each skirt-reinforcement member 730 is configured to restrict or prevent billowing, or radial motion, of the skirt 112 that spans across or extends over the respective side opening 118. More particularly, each skirt-reinforcement member 730 has a first end 732, a second end 734, and a length L therebetween. The skirt-reinforcement member 730, which includes the plurality of stitches 740, is directly attached to the skirt 112 along its length L. Stated another way, the skirt-reinforcement member 730 is directly attached to the skirt 112 along an unsupported portion of the skirt 112 that spans the respective side opening 118. As used herein, "unsupported portion of the skirt" refers to areas of the skirt 112 in which a surface of the skirt does not directly contact or abut against the valve support 104. The plurality of stitches 740 of the skirt-reinforcement member 730 extends or weaves through the material of the skirt 112 to ensure that the

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skirt-reinforcement member 730 applies tension to the skirt 112 and thereby prevents undesired billowing of the skirt material.

The first end 732 of the skirt-reinforcement member 730 is attached to a first edge 736 defining the respective side opening 118 and the second end 734 of the skirt-reinforcement member 730 is attached to a second or opposing edge 738 defining the respective side opening 118. Attaching the first and second ends 732, 734 of the skirt-reinforcement member 730 to the valve support 104 ensures that the skirt-reinforcement member 730 has sufficient rigidity to apply tension to the skirt 112 and prevent billowing. More particularly, the first and second ends 732, 734 of the skirt-reinforcement member 730 are attached to the valve support 104 such that the skirt-reinforcement member 730 has sufficient tension along its length L to minimize radial movement of the skirt 112 throughout the cardiac cycle.

As will be described in more detail herein, the first and second ends 732, 734 of the skirt-reinforcement member 730 may be attached to a crown 120 or a strut 122 of the valve support 104. Stated another way, the first and second edges 736, 738 defining the respective side opening 118 may be edges of a crown 120 or a strut 122 of the valve support. In the embodiment depicted in FIGS. 7A and 7B, the first end 732 of the skirt-reinforcement member 730 is circumferentially aligned with the second end 734 of the skirt-reinforcement member 730 such that the skirt-reinforcement member 732 extends substantially perpendicular relative to an axis LA of the valve support 104. The skirt-reinforcement member 730 extends over the side opening 118 in a center or middle thereof, such that the skirt-reinforcement member 730 spans across or extends over the side opening 118 at a maximum width MW thereof. The skirt-reinforcement member 730 spans across the side opening 118 from one node 121 to another node 121 of the valve support 104.

In the embodiment depicted in FIGS. 7A and 7B, adjacent skirt-reinforcement members 730 are circumferentially aligned such that the skirt-reinforcement members 730 collectively circumscribe the valve support 104. However, adjacent skirt-reinforcement members 730 are not required to be circumferentially aligned and further each side opening 118 is not required to include a skirt reinforcement member as described above.

Other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIGS. 8A and 8B, a first end 832 of a skirt-reinforcement member 830 is circumferentially offset from a second end 834 of the skirt-reinforcement member 830 such that the skirt-reinforcement member 830 is angled relative to the axis LA of the valve support 104. The first end 832 of the skirt-reinforcement member 830 is attached to a first edge 836 defining the respective side opening 118 and the second end 834 of the skirt-reinforcement member 830 is attached to a second or opposing edge 838 defining the respective side opening 118. The skirt-reinforcement member 830 spans diagonally across the side opening 118 from a top end of one node 121 to a bottom end of another node 121 of the valve support 104, or alternatively may span diagonally across the side opening 118 from a node 121 to a strut 122 of the valve support 104. In the embodiment depicted in FIGS. 8A and 8B, adjacent skirt-reinforcement members 830 have alternating orientations (i.e., diagonally downward and diagonally upward) such that the first end 832 of a skirt-reinforcement member 830 circumferentially aligns with a second end 834 of an adjacent skirt-reinforcement member 830 and the skirt-reinforcement members 830 collectively circumscribe the valve support 104 in a zig-zag

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configuration. However, adjacent skirt-reinforcement members 830 are not required to have alternating orientations and further each side opening 118 is not required to include a skirt reinforcement member as described above.

In another embodiment depicted in FIGS. 9A and 9B, a first end 932 of a skirt-reinforcement member 930 is aligned with a second end 934 of the skirt-reinforcement member 930 such that the skirt-reinforcement member 932 extends substantially perpendicular relative to the axis LA of the valve support 104 similar to the skirt-reinforcement member 730 described above. The first end 932 of the skirt-reinforcement member 930 is attached to a first edge 936 defining the respective side opening 118 and the second end 934 of the skirt-reinforcement member 930 is attached to a second or opposing edge 938 defining the respective side opening 118. However, in the embodiment of FIGS. 9A and 9B, the skirt-reinforcement member 930 extends over the side opening 118 in a lower region thereof, the lower region having width W2 that is less than the maximum width MW. The skirt-reinforcement member 930 thus spans across or extends over the side opening 118 at the width W2 thereof. The skirt-reinforcement member 930 spans across the side opening 118 from one node 121 to another node 121 of the valve support 104, or alternatively may span across the side opening 118 from one strut 122 to another strut 122 of the valve support 104. In the embodiment depicted in FIGS. 9A and 9B, adjacent skirt-reinforcement members 930 are circumferentially aligned such that the skirt-reinforcement members 930 collectively circumscribe the valve support 104. However, adjacent skirt-reinforcement members 930 are not required to be circumferentially aligned and further each side opening 118 is not required to include a skirt reinforcement member as described above.

Although the above embodiments depict the skirt-reinforcement members 730, 830, 930 having a generally linear or straight configuration within a side opening 118, other configurations of the skirt-reinforcement members are contemplated herein. For example, the skirt-reinforcement members 730, 830, 930 may have a wavy, sinusoidal, or zig-zag configuration along its length or may be configured as "T" or "X" within a side opening 118.

Turning to FIGS. 10A and 10B, a skirt-reinforcement member 1030 according to another aspect of the disclosure is illustrated. More particularly, FIG. 10A and FIG. 10B are side and perspective atrial end views, respectively, of the valve support 104 including a plurality of skirt-reinforcement members 1030. In the embodiment of FIGS. 10A and 10B, the valve support 104 includes a row of side openings 118 around a circumference thereof and a skirt-reinforcement member 1030 spans across or extends over each side opening 118 of the row of side openings 118 around the circumference of the valve support 104. However, it is not required that a skirt-reinforcement member 1030 be utilized on every side opening 118 of the row of side openings 118 around the circumference of the valve support 104. Rather, a skirt-reinforcement member 1030 may be utilized only on the side openings 118 having skirt material that may come into contact with leaflets 109 when the leaflets are opening and closing in situ. At least one side opening 118 must include a skirt-reinforcement member 1030 according to an aspect of the present disclosure. Each skirt-reinforcement member 1030 spans a side opening 118 from a first crown or strut edge defining the at least one side opening 118 to a second crown or strut edge defining the at least one side opening 118.

Each skirt-reinforcement member 1030 is similar to the skirt-reinforcement member 730 except that each skirt-

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reinforcement member **1030** further includes a strip **1042** of fabric or tissue in addition to a filament or strand **1031** of suture that forms a plurality of stitches **1040**. The strips **1042** are attached to the skirt **112** via the plurality of stitches **1040** and may be disposed on an outer surface or an inner surface of the skirt **112**. Thus, as compared to the skirt-reinforcement member **730** described above which includes a plurality of stitches **740** formed by suture alone, the skirt-reinforcement member **1030** further includes strips **1042** of tissue or fabric in addition to the plurality of stitches **1040** formed by suture and thus provides more robust support or reinforcement.

Similar to the skirt-reinforcement member **730**, each skirt-reinforcement member **1030** is configured to restrict or prevent billowing, or radial motion, of the skirt **112** that spans across or extends over the respective side opening **118**. More particularly, each skirt-reinforcement member **1030** has a first end **1032**, a second end **1034**, and a length *L* therebetween. The length *L* of the skirt-reinforcement member **1030**, which includes the plurality of stitches **1040** and one or more strips **1042** of tissue or fabric, is directly attached to the skirt **112** along its length *L*. Stated another way, the skirt-reinforcement member **1030** is directly attached to the skirt **112** along an unsupported portion of the skirt **112** that spans the respective side opening **118**. The plurality of stitches **1040** of the skirt-reinforcement member **1030** extends or weaves through the material of the skirt **112** and through the material of the strips **1042** of tissue or fabric, to ensure that the skirt-reinforcement member **1030** applies tension to the skirt **112** and thereby prevents undesired billowing of the skirt material.

The first end **1032** of the skirt-reinforcement member **1030** is formed by the strand **1031** of suture of the skirt-reinforcement member **1030**, and is attached to a first edge **1036** defining the respective side opening **118**. The second end **1034** of the skirt-reinforcement member **1030** is also formed by the strand **1031** of suture of the skirt-reinforcement member **1030**, and is attached to a second or opposing edge **1038** defining the respective side opening **118**. Attaching the first and second ends **1032**, **1034** of the skirt-reinforcement member **1030** to the valve support **104** ensures that the skirt-reinforcement member **1030** has sufficient rigidity to apply tension to the skirt **112** and prevent billowing. More particularly, the first and second ends **1032**, **1034** of the skirt-reinforcement member **1030** are attached to the valve support **104** such that the skirt-reinforcement member **1030** has sufficient tension along its length *L* to minimize radial movement of the skirt **112** throughout the cardiac cycle.

The first and second ends **1032**, **1034** of the skirt-reinforcement member **1030** may be attached to a crown **120** or a strut **122** of the valve support **104**. Stated another way, the first and second edges **1036**, **1038** defining the respective side opening **118** may be edges of a crown **120** or a strut **122** of the valve support. In the embodiment depicted in FIGS. **10A** and **10B**, the first end **1032** of the skirt-reinforcement member **1030** is aligned with the second end **1034** of the skirt-reinforcement member **1030** such that the skirt-reinforcement member **1032** extends substantially perpendicular relative to an axis *LA* of the valve support **104**. The skirt-reinforcement member **1030** extends over the side opening **118** in a center or middle thereof, such that the skirt-reinforcement member **1030** spans across or extends over the side opening **118** at a maximum width *MW* thereof. The skirt-reinforcement member **1030** spans across the side opening **118** from one node **121** to another node **121** of the valve support **104**.

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The strips **1042** may be a natural or biological material such as pericardium or another membranous tissue such as intestinal submucosa. Alternatively, the strips **1042** may be a low-porosity woven or knit fabric, such as polyester, Dacron fabric, or PTFE knit. Exemplary suture materials for the strand **1031** include but are not limited to a monofilament or plastic suture material, such as polypropylene.

The strips **1042** are relatively thin and have a thickness between 0.1 mm and 0.5 mm, and have a height (in the axial direction of the valve support **104**) between 0.2 mm up to the full height of the side opening **118**. The length of each strip **1042** is substantially the same as the length *L* of the strip-reinforcement member **1030** such that the strip **1042** extends substantially between the first end **1032** and the second end **1034** of the strip-reinforcement member **1030**.

In another embodiment, the strip **1042** of fabric or tissue may be attached to the skirt **112** with a different fixation mechanism as an alternative to the plurality of stitches **1040**. For example, in another embodiment, the strip **1042** of fabric or tissue may be attached to the skirt **112** using adhesive. When utilizing adhesive as a fixation mechanism, the strip **1042** of fabric or tissue is disposed onto the skirt in a taut configuration to ensure that the skirt-reinforcement member **1030** applies tension to the skirt **112** and thereby prevents undesired billowing of the skirt material. Further, the adhesive may retract as it cures to apply further tension to the skirt material. Adhesive as a fixation mechanism provides more predictable packing or folding of the valve support **104** when the valve support **104** is collapsed into the radially compressed or crimped configuration for delivery.

In the embodiment depicted in FIGS. **10A** and **10B**, adjacent skirt-reinforcement members **1030** are circumferentially aligned such that the skirt-reinforcement members **1030** collectively circumscribe the valve support **104**. However, adjacent skirt-reinforcement members **1030** are not required to be circumferentially aligned and further each side opening **118** is not required to include a skirt reinforcement member as described above.

Other placements of the skirt-reinforcement member **1030** are contemplated herein. For example, in the embodiment depicted in FIGS. **11A** and **11B**, a first end **1132** of a skirt-reinforcement member **1130** is offset from a second end **1134** of the skirt-reinforcement member **1130** such that the skirt-reinforcement member **1130** is angled relative to the axis *LA* of the valve support **104**. The first end **1132** of the skirt-reinforcement member **1130** is attached to a first edge **1136** defining the respective side opening **118** and the second end **1134** of the skirt-reinforcement member **1130** is attached to a second or opposing edge **1138** defining the respective side opening **118**. The skirt-reinforcement member **1130** spans diagonally across the side opening **118** from a top end of one node **121** to a bottom end of another node **121** of the valve support **104**, or alternatively may span diagonally across the side opening **118** from a node **121** to a strut **122** of the valve support **104**. In the embodiment depicted in FIGS. **11A** and **11B**, adjacent skirt-reinforcement members **1130** have alternating orientations (i.e., diagonally downward and diagonally upward) such that the first end **1132** of a skirt-reinforcement member **1130** circumferentially aligns with a second end **1134** of an adjacent skirt-reinforcement member **1130** and the skirt-reinforcement members **1130** collectively circumscribe the valve support **104** in a zig-zag configuration. However, adjacent skirt-reinforcement members **1130** are not required to have alternating orientations and further each side opening **118** is not required to include a skirt reinforcement member as described above.

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In another embodiment depicted in FIGS. 12A and 12B, a first end 1232 of a skirt-reinforcement member 1230 is aligned with a second end 1234 of the skirt-reinforcement member 1230 such that the skirt-reinforcement member 1232 extends substantially perpendicular relative to the axis LA of the valve support 104 similar to the skirt-reinforcement member 1030 described above. The first end 1232 of the skirt-reinforcement member 1230 is attached to a first edge 1236 defining the respective side opening 118 and the second end 1234 of the skirt-reinforcement member 1230 is attached to a second or opposing edge 1238 defining the respective side opening 118. However, in the embodiment of FIGS. 12A and 12B, the skirt-reinforcement member 1230 extends over the side opening 118 in a lower region thereof, the lower region having width W2 that is less than the maximum width MW. The skirt-reinforcement member 1230 thus spans across or extends over the side opening 118 at the width W2 thereof. The skirt-reinforcement member 1230 spans across the side opening 118 from one node 121 to another node 121 of the valve support 104, or alternatively may span across the side opening 118 from one strut 122 to another strut 122 of the valve support 104. In the embodiment depicted in FIGS. 12A and 12B, adjacent skirt-reinforcement members 1230 are circumferentially aligned such that the skirt-reinforcement members 1230 collectively circumscribe the valve support 104. However, adjacent skirt-reinforcement members 1230 are not required to be circumferentially aligned and further each side opening 118 is not required to include a skirt reinforcement member as described above.

Although the above embodiments depict the skirt-reinforcement members 1030, 1130, 1230 having a generally linear or straight configuration, other configurations of the skirt-reinforcement members are contemplated herein. For example, the skirt-reinforcement members 1030, 1130, 1230 may have a wavy, sinusoidal, or zig-zag configuration along its length or may be configured as "T" or "X" within the side opening 118.

Turning now to FIG. 13A and FIG. 13B, a skirt-reinforcement member 1330 according to another aspect of the disclosure is illustrated. More particularly, FIG. 13A is side view of the valve support 104 including the skirt-reinforcement member 1330 and FIG. 13B illustrates a top view of the skirt-reinforcement member 1330 removed from the valve support 104 for sake of illustration. In the embodiment of FIG. 13A and FIG. 13B, the skirt-reinforcement member 1330 is a continuous band or ring 1344 that extends 360° around an outer surface 105 of the valve support 104. Stated another way, the skirt-reinforcement member 1330 is circumferentially disposed around the perimeter of the valve support 104 and encircles or circumscribes the outer surface 105 of the valve support 104. The ring 1344 is an elastic material and a diameter D1 of the skirt-reinforcement member 1330 is approximately the same as a diameter D2 of the valve support 104 when the valve support 104 is in its radially expanded configuration.

Notably, the ring 1344 is directly attached to the skirt 112 along portions thereof that span across or extend over the side openings 118 of the valve support 104 but the ring 1344 is not directly attached to the valve support 104. Stated another way, when the ring 1344 and the valve support 104 meet or intersect, the ring 1344 overlaps or overlays the valve support 104 but is not otherwise attached thereto. The ring 1344 is however attached to the skirt 112 to apply tension to the skirt 112 and thereby restrict or prevent billowing of the skirt 112 that spans across or extends over the side openings 118 of the valve support 104. Thus, when

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the ring 1344 and the skirt 112 meet or intersect, the ring 1344 overlaps or overlays the skirt 112 and is further directly attached thereto. Since the ring 1344 is directly attached to the skirt 112, is formed from an elastic material, and has the diameter D that is approximately the same as the expanded diameter of the valve support 104, the skirt-reinforcement member 1330 has sufficient tension to minimize radial movement of the skirt 112 throughout the cardiac cycle. As such, the skirt-reinforcement member 1330 is configured to minimize radial movement of the skirt 112 and better protect the leaflets 109 from abrasion as well as prevent skirt degradation.

The ring 1344 may be attached to the skirt 112 via any suitable attachment mechanism, including but not limited to suture, adhesive, or welding. Suitable elastic materials for the ring 1344 include but are not limited to polymer materials such as polyurethane or silicone, as well as biological or natural materials such as pericardium or another membranous tissue such as intestinal submucosa. "Elastic" as used herein includes materials that may be stretched or elongated to fit over and around the outer surface 105 of the valve support 104, while also having sufficient resiliency to resume their original size/configuration and conform to the outer surface 105 of the valve support 104. The elastic material of the ring 1344 has sufficient resiliency to conform to and be secured over the outer surface 105 of the valve support 104, but does not exert an amount of force that would result in constriction or reduction of the inner diameter of the valve support 104.

As stated above, the skirt-reinforcement members 730, 830, 930, 1030, 1130, 1230, 1330 may be utilized on any transcatheter heart valve prosthesis that includes a skirt on a portion thereof. The skirt-reinforcement members 730, 830, 930, 1030, 1130, 1230, 1330 serve to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt and the leaflets. Depending on the configuration of the frame, the particular placement of the skirt-reinforcement members may vary. In an embodiment, the skirt-reinforcement members are positioned downstream of a margin of attachment (MOA) of the leaflets of the transcatheter heart valve prosthesis and are positioned over a side opening of the frame that includes skirt material over a portion thereof. The configuration of the frame and the configuration of the side opening of the frame is non-limiting and may vary, as exemplified in the following figures.

FIG. 14 depicts a perspective view of a transcatheter heart valve prosthesis 1400 in accordance with another aspect of the disclosure. The transcatheter heart valve prosthesis 1400 includes a stent or frame 1402 and a prosthetic valve component 1408 including at least one leaflet disposed within and secured to the frame 1402. The frame 1402 includes a plurality of crowns 1420 and a plurality of struts 1422 with each crown 1420 being formed between a pair of opposing struts 1422. Each crown 1420 is a curved segment or bend extending between opposing struts 1422. The frame 1402 is tubular, with the plurality of side openings 1418 being defined by edges of the plurality of crowns 1420 and the plurality of struts 1422. In an embodiment, the plurality of side openings 1418 may be substantially diamond-shaped.

The frame 1402 includes a skirt 1412 coupled to a surface thereof. More particularly, the skirt 1412 is coupled to an inner surface of the frame 1402 to line a portion thereof. Alternatively, the skirt 1412 may be coupled to an outer surface of the frame 1402 to enclose a portion thereof as would be known to one of ordinary skill in the art of

prosthetic valve construction. The skirt **1412** may be formed from the same materials described above with respect to the skirt **112**.

The prosthetic valve component **1408** of the transcatheter heart valve prosthesis **1400** is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. FIG. **14** illustrates an exemplary prosthetic valve component having three leaflets **1409**, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component **108**, when deployed in situ, the prosthetic valve component **1408** in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame **1402**.

The valve leaflets **1409** may be attached to the skirt **1412**. The skirt **1412** may billow during valve opening and closing in situ, and the leaflets **1409** may contact the skirt **1412**. Such billowing may undesirably result in contact between the skirt **1412** and the leaflets **1409** of the transcatheter heart valve prosthesis **1400**. If the leaflets **1409** of the transcatheter heart valve prosthesis **1400** contact the skirt **1412** during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt **1412** and the frame **1402** may further induce early skirt abrasion. Thus, the transcatheter heart valve prosthesis **1400** further includes a plurality of skirt-reinforcement members **1430**. As described above with respect to the skirt-reinforcement members **730** described above, the skirt-reinforcement members **1430** reinforce the material of the skirt **1412** that spans across the side openings **1418** of the frame **1402** to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt **1412** and the leaflets **1409**.

In the embodiment of FIG. **14**, the frame **1402** includes a row of side openings **1418** around a circumference thereof and a skirt-reinforcement member **1430** spans across or extends over each side opening **1418** of the row of side openings **1418** around the circumference of the frame **1402**. However, it is not required that a skirt-reinforcement member **1430** be utilized on every side opening **1418** of the row of side openings **1418** around the circumference of the frame **1402**. Rather, a skirt-reinforcement member **1430** may be utilized only on the side openings **1418** having skirt material that may come into contact with leaflets **1409** when the leaflets are opening and closing in situ. At least one side opening **1418** must include a skirt-reinforcement member **1430** according to an aspect of the present disclosure. Further, the placement of the skirt-reinforcement members **1430** is exemplary only and the skirt-reinforcement members **1430** may be disposed on side openings **1418** of different rows from that depicted on FIG. **14**.

FIG. **14A** is an enlarged view of a side opening **1418** of the frame **1402** that includes a skirt-reinforcement member **1430**. Each skirt-reinforcement member **1430** spans a side opening **1418** from a first crown or strut edge defining the at least one side opening **1418** to a second crown or strut edge defining the at least one side opening **1418**. In the embodiments of FIGS. **14-14A**, the skirt-reinforcement members **1430** are the same as the skirt-reinforcement members **730**. Each skirt-reinforcement member **1430** is a filament or strand **1431** of suture that forms a plurality of stitches **1440**. The skirt-reinforcement member **1430** is directly attached to the skirt **1412** along an unsupported portion of the skirt **1412** that spans the respective side opening **1418**. The plurality of stitches **1440** of the skirt-reinforcement member **1430**

extends or weaves through the material of the skirt **1412** to ensure that the skirt-reinforcement member **1430** applies tension to the skirt **1412** and thereby prevents undesired billowing of the skirt material. The first and second ends **1432**, **1434** of the skirt-reinforcement member **1430** are attached to the frame **1402** such that the skirt-reinforcement member **1430** has sufficient tension along its length to minimize radial movement of the skirt **1412** throughout the cardiac cycle.

In FIGS. **14** and **14A**, the skirt-reinforcement member **1430** extends in a horizontal orientation over the side opening **1418** in a center or middle thereof. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. **14B**, a skirt-reinforcement member **1430B** extends in a vertical orientation over the side opening **1418** in a center or middle thereof. A first end **1432B** of a skirt-reinforcement member **1430B** is longitudinally offset from a second end **1434B** of the skirt-reinforcement member **1430B** such that the skirt-reinforcement member **1430B** is parallel or substantially parallel relative to the axis **LA** of the frame. The first end **1432B** of the skirt-reinforcement member **1430B** is attached to a first crown **1420B** defining the respective side opening **1418** and the second end **1434B** of the skirt-reinforcement member **1430** is attached to a second or opposing crown **1420B** defining the respective side opening **1418** such that the skirt-reinforcement member **1430B** spans vertically across the side opening **1418**.

In another embodiment depicted in FIG. **14C**, a first end **1432C** of a skirt-reinforcement member **1430C** is circumferentially offset from a second end **1434C** of the skirt-reinforcement member **1430C** such that the skirt-reinforcement member **1430C** is angled relative to the axis **LA** of the frame. The first end **1432C** of the skirt-reinforcement member **1430C** is attached to a first edge **1436C** defining the respective side opening **1418** and the second end **1434C** of the skirt-reinforcement member **1430C** is attached to a second or opposing edge **1438C** defining the respective side opening **1418**. Adjacent skirt-reinforcement members **1430C** may have alternating orientations (i.e., diagonally downward and diagonally upward) such that the skirt-reinforcement members **1430C** collectively circumscribe the frame in a zig-zag configuration. However, adjacent skirt-reinforcement members **1430C** are not required to have alternating orientations and further each side opening **1418** is not required to include a skirt reinforcement member as described above.

Similar to the skirt-reinforcement members **1030**, **1130**, **1230** described above, the skirt-reinforcement members **1430**, **1430B**, **1430C** may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members **1430**, the transcatheter heart valve prosthesis **1400** may include a continuous band or ring that extends 360° around an outer surface of the frame **1402**. The ring would be attached to the skirt **1412** and may be the same as the skirt-reinforcement member **1330** described above. The ring would be positioned over the same row of side openings **1418** as shown with respect to the skirt-reinforcement members **1430** in FIG. **14**.

FIG. **15** depicts a perspective view of a transcatheter heart valve prosthesis **1500** in accordance with another aspect of the disclosure. The transcatheter heart valve prosthesis **1500** includes a stent or frame **1502** and a prosthetic valve component **1508** including at least one leaflet disposed within and secured to the frame **1502**. The frame **1502** includes a plurality of crowns **1520** and a plurality of struts

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1522 with each crown 1520 being formed between a pair of opposing struts 1522. Each crown 1520 is a curved segment or bend extending between opposing struts 1522. The frame 1502 is tubular, with the plurality of side openings 1518 being defined by edges of the plurality of crowns 1520 and the plurality of struts 1522. In an embodiment, the plurality of side openings 1518 may be substantially hexagonal-shaped.

The frame 1502 includes a skirt 1512 coupled to a surface thereof. More particularly, the skirt 1512 is coupled to an inner surface of the frame 1502 to line a portion thereof. Alternatively, the skirt 1512 may be coupled to an outer surface of the frame 1502 to enclose a portion thereof as would be known to one of ordinary skill in the art of prosthetic valve construction. The skirt 1512 may be formed from the same materials described above with respect to the skirt 112.

The prosthetic valve component 1508 of the transcatheter heart valve prosthesis 1500 is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. FIG. 15 illustrates an exemplary prosthetic valve component having three leaflets 1509, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component 108, when deployed in situ, the prosthetic valve component 1508 in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame 1502.

The valve leaflets 1509 may be attached to the skirt 1512. The skirt 1512 may billow during valve opening and closing in situ, and the leaflets 1509 may contact the skirt 1512. Such billowing may undesirably result in contact between the skirt 1512 and the leaflets 1509 of the transcatheter heart valve prosthesis 1500. If the leaflets 1509 of the transcatheter heart valve prosthesis 1500 contact the skirt 1512 during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt 1512 and the frame 1502 may further induce early skirt abrasion. Thus, the transcatheter heart valve prosthesis 1500 further includes a plurality of skirt-reinforcement members 1530. As described above with respect to the skirt-reinforcement members 730 described above, the skirt-reinforcement members 1530 reinforce the material of the skirt 1512 that spans across the side openings 1518 of the frame 1502 to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt 1512 and the leaflets 1509.

In the embodiment of FIG. 15, the frame 1502 includes a row of side openings 1518 around a circumference thereof and a skirt-reinforcement member 1530 spans across or extends over each side opening 1518 of the row of side openings 1518 around the circumference of the frame 1502. However, it is not required that a skirt-reinforcement member 1530 be utilized on every side opening 1518 of the row of side openings 1518 around the circumference of the frame 1502. Rather, a skirt-reinforcement member 1530 may be utilized only on the side openings 1518 having skirt material that may come into contact with leaflets 1509 when the leaflets are opening and closing in situ. At least one side opening 1518 must include a skirt-reinforcement member 1530 according to an aspect of the present disclosure. Further, the placement of the skirt-reinforcement members 1530 is exemplary only and the skirt-reinforcement members 1530 may be disposed on side openings 1518 of different rows from that depicted on FIG. 15.

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FIG. 15A is an enlarged view of a side opening 1518 of the frame 1502 that includes a skirt-reinforcement member 1530. Each skirt-reinforcement member 1530 spans a side opening 1518 from a first crown or strut edge defining the at least one side opening 1518 to a second crown or strut edge defining the at least one side opening 1518. In the embodiments of FIGS. 15-15A, the skirt-reinforcement members 1530 are the same as the skirt-reinforcement members 730. Each skirt-reinforcement member 1530 is a filament or strand 1531 of suture that forms a plurality of stitches 1540. The skirt-reinforcement member 1530 is directly attached to the skirt 1512 along an unsupported portion of the skirt 1512 that spans the respective side opening 1518. A plurality of stitches 1540 of the skirt-reinforcement member 1530 extends or weaves through the material of the skirt 1512 to ensure that the skirt-reinforcement member 1530 applies tension to the skirt 1512 and thereby prevents undesired billowing of the skirt material. The first and second ends 1532, 1534 of the skirt-reinforcement member 1530 are attached to the frame 1502 such that the skirt-reinforcement member 1530 has sufficient tension along its length to minimize radial movement of the skirt 1512 throughout the cardiac cycle.

In FIGS. 15 and 15A, the skirt-reinforcement member 1530 extends in a horizontal orientation over the side opening 1518 in a center or middle thereof. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. 15B, a skirt-reinforcement member 1530B extends in a vertical orientation over the side opening 1518 in a center or middle thereof. A first end 1532B of a skirt-reinforcement member 1530B is longitudinally offset from a second end 1534B of the skirt-reinforcement member 1530B such that the skirt-reinforcement member 1530B is parallel or substantially parallel relative to the axis LA of the frame. The first end 1532B of the skirt-reinforcement member 1530B is attached to a first crown 1520B defining the respective side opening 1518 and the second end 1534B of the skirt-reinforcement member 1530 is attached to a second or opposing crown 1520B defining the respective side opening 1518 such that the skirt-reinforcement member 1530B spans vertically across the side opening 1518.

In another embodiment depicted in FIG. 15C, a first end 1532C of a skirt-reinforcement member 1530C is circumferentially offset from a second end 1534C of the skirt-reinforcement member 1530C such that the skirt-reinforcement member 1530C is angled relative to the axis LA of the frame. The first end 1532C of the skirt-reinforcement member 1530C is attached to a first edge 1536C defining the respective side opening 1518 and the second end 1534C of the skirt-reinforcement member 1530C is attached to a second or opposing edge 1538C defining the respective side opening 1518. Adjacent skirt-reinforcement members 1530C may have alternating orientations (i.e., diagonally downward and diagonally upward) such that the skirt-reinforcement members 1530C collectively circumscribe the frame in a zig-zag configuration. However, adjacent skirt-reinforcement members 1530C are not required to have alternating orientations and further each side opening 1518 is not required to include a skirt reinforcement member as described above.

Similar to the skirt-reinforcement members 1030, 1130, 1230 described above, the skirt-reinforcement members 1530, 1530B, 1530C may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members 1530, the transcath-

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eter heart valve prosthesis **1500** may include a continuous band or ring that extends 360° around an outer surface of the frame **1502**. The ring would be attached to the skirt **1512** and may be the same as the skirt-reinforcement member **1330** described above. The ring would be positioned over the same row of side openings **1518** as shown with respect to the skirt-reinforcement members **1530** in FIG. 15.

FIG. 16 depicts a perspective view of a transcatheter heart valve prosthesis **1600** in accordance with another aspect of the disclosure. The transcatheter heart valve prosthesis **1600** includes a stent or frame **1602** and a prosthetic valve component **1608** including at least one leaflet disposed within and secured to the frame **1602**. The frame **1602** includes a plurality of crowns **1620** and a plurality of struts **1622** with each crown **1620** being formed between a pair of opposing struts **1622**. Each crown **1620** is a curved segment or bend extending between opposing struts **1622**. The frame **1602** is tubular, with the plurality of side openings **1618** being defined by edges of the plurality of crowns **1620** and the plurality of struts **1622**. In an embodiment, the plurality of side openings **1618** may be substantially diamond-shaped.

The frame **1602** includes a skirt **1612** coupled to a surface thereof. More particularly, the skirt **1612** is coupled to an inner surface of the frame **1602** to line a portion thereof. Alternatively, the skirt **1612** may be coupled to an outer surface of the frame **1602** to enclose a portion thereof as would be known to one of ordinary skill in the art of prosthetic valve construction. The skirt **1612** may be formed from the same materials described above with respect to the skirt **112**.

The prosthetic valve component **1608** of the transcatheter heart valve prosthesis **1600** is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. FIG. 16 illustrates an exemplary prosthetic valve component having three leaflets **1609**, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component **108**, when deployed in situ, the prosthetic valve component **1608** in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame **1602**.

The valve leaflets **1609** may be attached to the skirt **1612**. The skirt **1612** may billow during valve opening and closing in situ, and the leaflets **1609** may contact the skirt **1612**. Such billowing may undesirably result in contact between the skirt **1612** and the leaflets **1609** of the transcatheter heart valve prosthesis **1600**. If the leaflets **1609** of the transcatheter heart valve prosthesis **1600** contact the skirt **1612** during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt **1612** and the frame **1602** may further induce early skirt abrasion. Thus, the transcatheter heart valve prosthesis **1600** further includes a plurality of skirt-reinforcement members **1630**. As described above with respect to the skirt-reinforcement members **730** described above, the skirt-reinforcement members **1630** reinforce the material of the skirt **1612** that spans across the side openings **1618** of the frame **1602** to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt **1612** and the leaflets **1609**.

In the embodiment of FIG. 16, the frame **1602** includes exactly three skirt-reinforcement members **1630** (although two skirt-reinforcement members are obscured from view). The skirt-reinforcement members **1630** are disposed at the

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leaflet commissures and are spaced at substantially equal intervals around a circumference of the frame **102**. However, the placement of the skirt-reinforcement members **1630** is exemplary only and the skirt-reinforcement members **1630** may be disposed at different side openings **1618** from those depicted on FIG. 16.

FIG. 16A is an enlarged view of a side opening **1618** of the frame **1602** that includes a skirt-reinforcement member **1630**. Each skirt-reinforcement member **1630** spans a side opening **1618** from a first crown or strut edge defining the at least one side opening **1618** to a second crown or strut edge defining the at least one side opening **1618**. In the embodiments of FIGS. 16-16A, the skirt-reinforcement members **1630** are the same as the skirt-reinforcement members **730**. Each skirt-reinforcement member **1630** is a filament or strand **1631** of suture that forms a plurality of stitches **1640**. The skirt-reinforcement member **1630** is directly attached to the skirt **1612** along an unsupported portion of the skirt **1612** that spans the respective side opening **1618**. A plurality of stitches **1640** of the skirt-reinforcement member **1630** extends or weaves through the material of the skirt **1612** to ensure that the skirt-reinforcement member **1630** applies tension to the skirt **1612** and thereby prevents undesired billowing of the skirt material. The first and second ends **1632**, **1634** of the skirt-reinforcement member **1630** are attached to the frame **1602** such that the skirt-reinforcement member **1630** has sufficient tension along its length to minimize radial movement of the skirt **1612** throughout the cardiac cycle.

In FIGS. 16 and 16A, the skirt-reinforcement member **1630** extends in a horizontal orientation over the side opening **1618** in a center or middle thereof. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. 16B, a skirt-reinforcement member **1630B** extends in a vertical orientation over the side opening **1618** in a center or middle thereof. A first end **1632B** of a skirt-reinforcement member **1630B** is longitudinally offset from a second end **1634B** of the skirt-reinforcement member **1630B** such that the skirt-reinforcement member **1630B** is parallel or substantially parallel relative to the axis LA of the frame. The first end **1632B** of the skirt-reinforcement member **1630B** is attached to a first crown **1620B** defining the respective side opening **1618** and the second end **1634B** of the skirt-reinforcement member **1630** is attached to a second or opposing crown **1620B** defining the respective side opening **1618** such that the skirt-reinforcement member **1630B** spans vertically across the side opening **1618**.

In another embodiment depicted in FIG. 16C, a first end **1632C** of a skirt-reinforcement member **1630C** is circumferentially offset from a second end **1634C** of the skirt-reinforcement member **1630C** such that the skirt-reinforcement member **1630C** is angled relative to the axis LA of the frame. The first end **1632C** of the skirt-reinforcement member **1630C** is attached to a first edge **1636C** defining the respective side opening **1618** and the second end **1634C** of the skirt-reinforcement member **1630C** is attached to a second or opposing edge **1638C** defining the respective side opening **1618**.

Similar to the skirt-reinforcement members **1030**, **1130**, **1230** described above, the skirt-reinforcement members **1630**, **1630B**, **1630C** may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members **1630**, the transcatheter heart valve prosthesis **1600** may include a continuous band or ring that extends 360° around an outer surface of the

frame **1602**. The ring would be attached to the skirt **1612** and may be the same as the skirt-reinforcement member **1330** described above.

FIG. **17** depicts a perspective view of a transcatheter heart valve prosthesis **1700** in accordance with another aspect of the disclosure. The transcatheter heart valve prosthesis **1700** includes a stent or frame **1702** and a prosthetic valve component **1708** including at least one leaflet disposed within and secured to the frame **1702**. The frame **1702** includes a plurality of crowns **1720** and a plurality of struts **1722** with each crown **1720** being formed between a pair of opposing struts **1722**. Each crown **1720** is a curved segment or bend extending between opposing struts **1722**. The frame **1702** is tubular, with the plurality of side openings **1718** being defined by edges of the plurality of crowns **1720** and the plurality of struts **1722**. In an embodiment, the plurality of side openings **1718** may be substantially heart-shaped, with each side opening **1718** being formed by six struts **1722** and six crowns **1720**.

The frame **1702** includes a skirt **1712** coupled to a surface thereof. More particularly, the skirt **1712** is coupled to an inner surface of the frame **1702** to line a portion thereof. Alternatively, the skirt **1712** may be coupled to an outer surface of the frame **1702** to enclose a portion thereof as would be known to one of ordinary skill in the art of prosthetic valve construction. The skirt **1712** may be formed from the same materials described above with respect to the skirt **112**.

The prosthetic valve component **1708** of the transcatheter heart valve prosthesis **1700** is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. FIG. **17** illustrates an exemplary prosthetic valve component having three leaflets **1709**, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component **108**, when deployed in situ, the prosthetic valve component **1708** in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame **1702**.

The valve leaflets **1709** may be attached to the skirt **1712**. The skirt **1712** may billow during valve opening and closing in situ, and the leaflets **1709** may contact the skirt **1712**. Such billowing may undesirably result in contact between the skirt **1712** and the leaflets **1709** of the transcatheter heart valve prosthesis **1700**. If the leaflets **1709** of the transcatheter heart valve prosthesis **1700** contact the skirt **1712** during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt **1712** and the frame **1702** may further induce early skirt abrasion. Thus, the transcatheter heart valve prosthesis **1700** further includes a plurality of skirt-reinforcement members **1730**. As described above with respect to the skirt-reinforcement members **730** described above, the skirt-reinforcement members **1730** reinforce the material of the skirt **1712** that spans across the side openings **1718** of the frame **1702** to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt **1712** and the leaflets **1709**.

In the embodiment of FIG. **17**, the frame **1702** includes a row of side openings **1718** around a circumference thereof and a skirt-reinforcement member **1730** spans across or extends over each side opening **1718** of the row of side openings **1718** around the circumference of the frame **1702**. However, it is not required that a skirt-reinforcement member **1730** be utilized on every side opening **1718** of the row

of side openings **1718** around the circumference of the frame **1702**. Rather, a skirt-reinforcement member **1730** may be utilized only on the side openings **1718** having skirt material that may come into contact with leaflets **1709** when the leaflets are opening and closing in situ. At least one side opening **1718** must include a skirt-reinforcement member **1730** according to an aspect of the present disclosure. Further, the placement of the skirt-reinforcement members **1730** is exemplary only and the skirt-reinforcement members **1730** may be disposed on side openings **1718** of different rows from that depicted on FIG. **17**.

FIG. **17A** is an enlarged view of a side opening **1718** of the frame **1702** that includes a skirt-reinforcement member **1730**. Each skirt-reinforcement member **1730** spans a side opening **1718** from a first crown or strut edge defining the at least one side opening **1718** to a second crown or strut edge defining the at least one side opening **1718**. In the embodiments of FIGS. **17-17A**, the skirt-reinforcement members **1730** are the same as the skirt-reinforcement members **730**. Each skirt-reinforcement member **1730** is a filament or strand **1731** of suture that forms a plurality of stitches **1740**. The skirt-reinforcement member **1730** is directly attached to the skirt **1712** along an unsupported portion of the skirt **1712** that spans the respective side opening **1718**. A plurality of stitches **1740** of the skirt-reinforcement member **1730** extends or weaves through the material of the skirt **1712** to ensure that the skirt-reinforcement member **1730** applies tension to the skirt **1712** and thereby prevents undesired billowing of the skirt material. The first and second ends **1732**, **1734** of the skirt-reinforcement member **1730** are attached to the frame **1702** such that the skirt-reinforcement member **1730** has sufficient tension along its length to minimize radial movement of the skirt **1712** throughout the cardiac cycle.

In FIGS. **17** and **17A**, the skirt-reinforcement member **1730** extends in a horizontal orientation over the side opening **1718**. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. **17B**, a skirt-reinforcement member **1730B** extends in a vertical orientation over the side opening **1718** in a center or middle thereof. A first end **1732B** of a skirt-reinforcement member **1730B** is longitudinally offset from a second end **1734B** of the skirt-reinforcement member **1730B** such that the skirt-reinforcement member **1730B** is parallel or substantially parallel relative to the axis **LA** of the frame. The first end **1732B** of the skirt-reinforcement member **1730B** is attached to a first crown **1720B** defining the respective side opening **1718** and the second end **1734B** of the skirt-reinforcement member **1730** is attached to a second or opposing crown **1720B** defining the respective side opening **1718** such that the skirt-reinforcement member **1730B** spans vertically across the side opening **1718**.

In another embodiment depicted in FIG. **17C**, a first end **1732C** of a skirt-reinforcement member **1730C** is circumferentially offset from a second end **1734C** of the skirt-reinforcement member **1730C** such that the skirt-reinforcement member **1730C** is angled relative to the axis **LA** of the frame. The first end **1732C** of the skirt-reinforcement member **1730C** is attached to a first edge **1736C** defining the respective side opening **1718** and the second end **1734C** of the skirt-reinforcement member **1730C** is attached to a second or opposing edge **1738C** defining the respective side opening **1718**. Adjacent skirt-reinforcement members **1730C** may have alternating orientations (i.e., diagonally downward and diagonally upward) such that the skirt-reinforcement members **1730C** collectively circumscribe the

frame in a zig-zag configuration. However, adjacent skirt-reinforcement members **1730C** are not required to have alternating orientations and further each side opening **1718** is not required to include a skirt reinforcement member as described above.

In another embodiment depicted in FIG. **17D**, two skirt-reinforcement members **1730D₁**, **1730D₂** extend in a horizontal orientation over a single side opening **1718**. Each skirt-reinforcement member **1730D₁**, **1730D₂** spans the side opening **1718** from a first crown or strut edge defining the side opening **1718** to a second crown or strut edge defining the at least one side opening **1718**. In the depicted embodiment, each skirt-reinforcement member **1730D₁**, **1730D₂** has a first end attached to a first crown **1720** defining the respective side opening **1718** and a second end attached to a second or opposing crown **1720** defining the respective side opening **1718**. In the depicted embodiment, each skirt-reinforcement member **1730D₁**, **1730D₂** are aligned with each other and have the same orientation, but the placement of the skirt-reinforcement members **1730D₁**, **1730D₂** are exemplary only and the skirt-reinforcement members **1730D₁**, **1730D₂** may be disposed in different orientations from that depicted on FIG. **17**.

Similar to the skirt-reinforcement members **1030**, **1130**, **1230** described above, the skirt-reinforcement members **1730**, **1730B**, **1730C** may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members **1730**, the transcatheter heart valve prosthesis **1700** may include a continuous band or ring that extends 360° around an outer surface of the frame **1702**. The ring would be attached to the skirt **1712** and may be the same as the skirt-reinforcement member **1330** described above. The ring would be positioned over the same row of side openings **1718** as shown with respect to the skirt-reinforcement members **1730** in FIG. **17**.

FIG. **18** depicts a perspective view of a transcatheter heart valve prosthesis **1800** in accordance with another aspect of the disclosure. The transcatheter heart valve prosthesis **1800** includes a stent or frame **1802** and a prosthetic valve component **1808** including at least one leaflet disposed within and secured to the frame **1802**. The frame **1802** includes a plurality of crowns **1820** and a plurality of struts **1822** with each crown **1820** being formed between a pair of opposing struts **1822**. Each crown **1820** is a curved segment or bend extending between opposing struts **1822**. The frame **1802** is tubular, with the plurality of side openings **1818** being defined by edges of the plurality of crowns **1820** and the plurality of struts **1822**. The frame **1802** includes five rows of side openings **1818**, labeled as rows **r₁**, **r₂**, **r₃**, **r₄**, **r₅** on FIG. **18**. In an embodiment, the plurality of side openings **1818** on each of rows **r₃**, **r₄**, **r₅**, may be substantially diamond-shaped, while the side openings **1818** on row **r₂** are each defined by five struts **1822** and the side openings **1818** on row **r₁** are each defined by six struts **1822**.

The frame **1802** includes a skirt **1812** coupled to a surface thereof. More particularly, the skirt **1812** is coupled to an inner surface of the frame **1802** to line a portion thereof. Alternatively, the skirt **1812** may be coupled to an outer surface of the frame **1802** to enclose a portion thereof as would be known to one of ordinary skill in the art of prosthetic valve construction. The skirt **1812** may be formed from the same materials described above with respect to the skirt **112**.

The prosthetic valve component **1808** of the transcatheter heart valve prosthesis **1800** is capable of regulating flow therethrough via valve leaflets that may form a replacement

valve. FIG. **18** illustrates an exemplary prosthetic valve component having three leaflets **1809**, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component **108**, when deployed in situ, the prosthetic valve component **1808** in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame **1802**.

The valve leaflets **1809** may be attached to the skirt **1812**. The skirt **1812** may billow during valve opening and closing in situ, and the leaflets **1809** may contact the skirt **1812**. Such billowing may undesirably result in contact between the skirt **1812** and the leaflets **1809** of the transcatheter heart valve prosthesis **1800**. If the leaflets **1809** of the transcatheter heart valve prosthesis **1800** contact the skirt **1812** during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt **1812** and the frame **1802** may further induce early skirt abrasion. Thus, the transcatheter heart valve prosthesis **1800** further includes a plurality of skirt-reinforcement members **1830**. As described above with respect to the skirt-reinforcement members **730** described above, the skirt-reinforcement members **1830** reinforce the material of the skirt **1812** that spans across the side openings **1818** of the frame **1802** to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt **1812** and the leaflets **1809**.

In the embodiment of FIG. **18**, a skirt-reinforcement member **1830** spans across or extends over each side opening **1818** of row **r₂** of side openings **1818** around the circumference of the frame **1802** and as well as each side opening **1818** of row **r₄** of side openings **1818** around the circumference of the frame **1802**. However, it is not required that a skirt-reinforcement member **1830** be utilized on every side opening **1818** of the rows **r₂**, **r₄** of the side openings **1818** around the circumference of the frame **1802**. Rather, a skirt-reinforcement member **1830** may be utilized only on the side openings **1818** having skirt material that may come into contact with leaflets **1809** when the leaflets are opening and closing in situ. At least one side opening **1818** must include a skirt-reinforcement member **1830** according to an aspect of the present disclosure. Further, the placement of the skirt-reinforcement members **1830** on rows **r₂**, **r₄** of side openings **1818** is exemplary only and the skirt-reinforcement members **1830** may be disposed on the side openings **1818** of different rows from those depicted on FIG. **18**.

FIG. **18A** is an enlarged view of a side opening **1818** of the frame **1802** that includes a skirt-reinforcement member **1830**. Each skirt-reinforcement member **1830** spans a side opening **1818** from a first crown or strut edge defining the at least one side opening **1818** to a second crown or strut edge defining the at least one side opening **1818**. In the embodiments of FIGS. **18-18A**, the skirt-reinforcement members **1830** are the same as the skirt-reinforcement members **730**. Each skirt-reinforcement member **1830** is a filament or strand **1831** of suture that forms a plurality of stitches **1840**. The skirt-reinforcement member **1830** is directly attached to the skirt **1812** along an unsupported portion of the skirt **1812** that spans the respective side opening **1818**. A plurality of stitches **1840** of the skirt-reinforcement member **1830** extends or weaves through the material of the skirt **1812** to ensure that the skirt-reinforcement member **1830** applies tension to the skirt **1812** and thereby prevents undesired billowing of the skirt material. The first and second ends **1832**, **1834** of the skirt-reinforcement member **1830** are

attached to the frame **1802** such that the skirt-reinforcement member **1830** has sufficient tension along its length to minimize radial movement of the skirt **1812** throughout the cardiac cycle.

In FIGS. **18** and **18A**, the skirt-reinforcement member **1830** extends in a horizontal orientation over the side opening **1818** in a center or middle thereof. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. **18B**, a skirt-reinforcement member **1830B** extends in a vertical orientation over the side opening **1818** in a center or middle thereof. A first end **1832B** of a skirt-reinforcement member **1830B** is longitudinally offset from a second end **1834B** of the skirt-reinforcement member **1830B** such that the skirt-reinforcement member **1830B** is parallel or substantially parallel relative to the axis **LA** of the frame. The first end **1832B** of the skirt-reinforcement member **1830B** is attached to a first crown **1820B** defining the respective side opening **1818** and the second end **1834B** of the skirt-reinforcement member **1830** is attached to a second or opposing crown **1820B** defining the respective side opening **1818** such that the skirt-reinforcement member **1830B** spans vertically across the side opening **1818**.

In another embodiment depicted in FIG. **18C**, a first end **1832C** of a skirt-reinforcement member **1830C** is circumferentially offset from a second end **1834C** of the skirt-reinforcement member **1830C** such that the skirt-reinforcement member **1830C** is angled relative to the axis **LA** of the frame. The first end **1832C** of the skirt-reinforcement member **1830C** is attached to a first crown **1820**, or a first edge, defining the respective side opening **1818** and the second end **1834C** of the skirt-reinforcement member **1830C** is attached to a second or opposing crown **1820**, or a second or opposing edge, defining the respective side opening **1818**. Adjacent skirt-reinforcement members **1830C** may have alternating orientations (i.e., diagonally downward and diagonally upward) such that the skirt-reinforcement members **1830C** collectively circumscribe the frame in a zig-zag configuration. However, adjacent skirt-reinforcement members **1830C** are not required to have alternating orientations and further each side opening **1818** is not required to include a skirt reinforcement member as described above.

Similar to the skirt-reinforcement members **1030**, **1130**, **1230** described above, the skirt-reinforcement members **1830**, **1830B**, **1830C** may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members **1830**, the transcatheter heart valve prosthesis **1800** may include a continuous band or ring that extends 360° around an outer surface of the frame **1802**. The ring would be attached to the skirt **1812** and may be the same as the skirt-reinforcement member **1330** described above. The ring may be positioned over the same rows of side openings **1818** as shown with respect to the skirt-reinforcement members **1830** in FIG. **18**.

Although embodiments above depict various transcatheter valve prostheses having at least one skirt-reinforcement member, the skirt-reinforcement members described herein may be applied to a docking prosthesis that is configured to receive a valve prosthesis therein. Such docking prostheses may include a skirt on a portion thereof that may come into contact with a leaflet of the valve prosthesis received therein when the leaflets are opening and closing in situ. For example, FIG. **19** depicts a perspective view of a docking prosthesis **1900** in accordance with another aspect of the disclosure. The docking prosthesis **1900** includes a stent or frame **1902**. The frame **1902** includes a plurality of crowns

1920 and a plurality of struts **1922** with each crown **1920** being formed between a pair of opposing struts **1922**. Each crown **1920** is a curved segment or bend extending between opposing struts **1922**. The frame **1902** is tubular, with the plurality of side openings **1918** being defined by edges of the plurality of crowns **1920** and the plurality of struts **1922**. In an embodiment, the plurality of side openings **1918** may be substantially diamond-shaped.

The frame **1902** includes a skirt **1912** coupled to a surface thereof. More particularly, the skirt **1912** is coupled to an inner surface of the frame **1902** to line a portion thereof. Alternatively, the skirt **1912** may be coupled to an outer surface of the frame **1902** to enclose a portion thereof. The skirt **1912** may be formed from the same materials described above with respect to the skirt **112**.

With reference to FIG. **20**, the docking prosthesis **1900** is configured to receive a valve prosthesis **2000** therein. The valve prosthesis **2000** includes a frame **2002** and a prosthetic valve component **2008** including at least one leaflet disposed within and secured to the frame **2002**. The prosthetic valve component **2008** of the valve prosthesis **2000** is capable of regulating flow therethrough via valve leaflets that may form a replacement valve. The prosthetic valve component **2008** may have three leaflets **2009**, although a single leaflet or bicuspid leaflet configuration may alternatively be used in embodiments hereof. As described above with respect to the prosthetic valve component **108**, when deployed in situ, the prosthetic valve component **2008** in a closed state is configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame **2002**. The valve leaflets **2009** may be attached to the frame **2002**.

The skirt **1912** of the docking prosthesis **1900** may billow during opening and closing of the leaflets **2009** in situ, and the leaflets **2009** may contact the skirt **1912**. Such billowing may undesirably result in contact between the skirt **1912** of the docking prosthesis **1900** and the leaflets **2009** of the valve prosthesis **2000**. If the leaflets **2009** contact the skirt **1912** during opening and closing, such contact may cause early leaflet tissue abrasion as well as early skirt abrasion. Additionally, the greater relative motion between the skirt **1912** and the frame **1902** may further induce early skirt abrasion. Thus, the docking prosthesis **1900** further includes a plurality of skirt-reinforcement members **1930**. As described above with respect to the skirt-reinforcement members **730** described above, the skirt-reinforcement members **1930** reinforce the material of the skirt **1912** that spans across the side openings **1918** of the frame **1902** to limit the radial motion or billowing of the skirt material, thereby minimizing risk of damage to both the skirt **1912** and the leaflets **2009**.

In the embodiment of FIG. **19**, the frame **1902** includes a row of side openings **1918** around a circumference thereof and a skirt-reinforcement member **1930** spans across or extends over each side opening **1918** of the row of side openings **1918** around the circumference of the frame **1902**. However, it is not required that a skirt-reinforcement member **1930** be utilized on every side opening **1918** of the row of side openings **1918** around the circumference of the frame **1902**. Rather, a skirt-reinforcement member **1930** may be utilized only on the side openings **1918** having skirt material that may come into contact with leaflets **2009** when the leaflets are opening and closing in situ. At least one side opening **1918** must include a skirt-reinforcement member **1930** according to an aspect of the present disclosure. Further, the placement of the skirt-reinforcement members **1930** is exemplary only and the skirt-reinforcement mem-

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bers 1930 may be disposed on side openings 1918 of different rows from that depicted on FIG. 19.

FIG. 19A is an enlarged view of a side opening 1918 of the frame 1902 that includes a skirt-reinforcement member 1930. Each skirt-reinforcement member 1930 spans a side opening 1918 from a first crown or strut edge defining the at least one side opening 1918 to a second crown or strut edge defining the at least one side opening 1918. In the embodiments of FIGS. 19-19A, the skirt-reinforcement members 1930 are the same as the skirt-reinforcement members 730. Each skirt-reinforcement member 1930 is a filament or strand 1931 of suture that forms a plurality of stitches 1940. The skirt-reinforcement member 1930 is directly attached to the skirt 1912 along an unsupported portion of the skirt 1912 that spans the respective side opening 1918. A plurality of stitches 1940 of the skirt-reinforcement member 1930 extends or weaves through the material of the skirt 1912 to ensure that the skirt-reinforcement member 1930 applies tension to the skirt 1912 and thereby prevents undesired billowing of the skirt material. The first and second ends 1932, 1934 of the skirt-reinforcement member 1930 are attached to the frame 1902 such that the skirt-reinforcement member 1930 has sufficient tension along its length to minimize radial movement of the skirt 1912 throughout the cardiac cycle.

In FIGS. 19 and 19A, the skirt-reinforcement member 1930 extends in a horizontal orientation over the side opening 1918 in a center or middle thereof. However, other placements of the skirt-reinforcement members are contemplated herein. For example, in the embodiment depicted in FIG. 19B, a skirt-reinforcement member 1930B extends in a vertical orientation over the side opening 1918 in a center or middle thereof. A first end 1932B of a skirt-reinforcement member 1930B is longitudinally offset from a second end 1934B of the skirt-reinforcement member 1930B such that the skirt-reinforcement member 1930B is parallel or substantially parallel relative to the axis LA of the frame. The first end 1932B of the skirt-reinforcement member 1930B is attached to a first crown 1920B defining the respective side opening 1918 and the second end 1934B of the skirt-reinforcement member 1930 is attached to a second or opposing crown 1920B defining the respective side opening 1918 such that the skirt-reinforcement member 1930B spans vertically across the side opening 1918.

In another embodiment depicted in FIG. 19C, a first end 1932C of a skirt-reinforcement member 1930C is circumferentially offset from a second end 1934C of the skirt-reinforcement member 1930C such that the skirt-reinforcement member 1930C is angled relative to the axis LA of the frame. The first end 1932C of the skirt-reinforcement member 1930C is attached to a first edge 1936C defining the respective side opening 1918 and the second end 1934C of the skirt-reinforcement member 1930C is attached to a second or opposing edge 1938C defining the respective side opening 1918. Adjacent skirt-reinforcement members 1930C may have alternating orientations (i.e., diagonally downward and diagonally upward) such that the skirt-reinforcement members 1930C collectively circumscribe the frame in a zig-zag configuration. However, adjacent skirt-reinforcement members 1930C are not required to have alternating orientations and further each side opening 1918 is not required to include a skirt reinforcement member as described above.

Similar to the skirt-reinforcement members 1030, 1130, 1230 described above, the skirt-reinforcement members 1930, 1930B, 1930C may include a strip of fabric or tissue in addition to a filament or strand of suture that forms a

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plurality of stitches. In addition, an alternative to the plurality of skirt-reinforcement members 1930, the docking prosthesis 1900 may include a continuous band or ring that extends 360° around an outer surface of the frame 1902. The ring would be attached to the skirt 1912 and may be the same as the skirt-reinforcement member 1330 described above. The ring would be positioned over the same row of side openings 1918 as shown with respect to the skirt-reinforcement members 1930 in FIG. 19.

It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

What is claimed is:

1. A prosthesis having a radially expanded configuration and a radially compressed configuration, the prosthesis comprising:

a frame including a plurality of crowns and a plurality of struts with each crown being formed between a pair of opposing struts, wherein a plurality of side openings is defined by edges of the plurality of crowns and the plurality of struts;

a skirt coupled to a surface of the frame, wherein the skirt extends over at least one side opening of the plurality of side openings of the frame;

a prosthetic valve component including a plurality of leaflets disposed within and secured to the frame, the prosthetic valve being configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame, wherein a margin of attachment is formed at the junction of the leaflets to the skirt; and

a skirt-reinforcement member that spans the at least one side opening of the plurality of side openings of the frame from a first crown or strut edge defining the at least one side opening to a second crown or strut edge defining the at least one side opening, wherein the skirt-reinforcement member is longitudinally spaced apart from the margin of attachment, the skirt-reinforcement member having a first end, a second end, and a length therebetween, wherein the first end of the skirt-reinforcement member is attached to the first crown or strut edge and the second end of the skirt-reinforcement member is attached to the second crown or strut edge, and wherein the skirt-reinforcement member is attached to the skirt along an unsupported portion of the skirt that spans the at least one side opening.

2. The prosthesis of claim 1, wherein the skirt-reinforcement member includes a suture forming a plurality of stitches.

3. The prosthesis of claim 1, wherein the skirt-reinforcement member includes tissue.

4. The prosthesis of claim 1, wherein the skirt-reinforcement member includes fabric.

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5. The prosthesis of claim 1, wherein an end of the frame includes a row of side openings around a circumference of the frame, the row including between six and nine side openings.

6. The prosthesis of claim 1, wherein the first end of the skirt-reinforcement member is circumferentially offset from the second end of the skirt-reinforcement member such that the skirt-reinforcement member is angled relative to an axis of the frame.

7. The prosthesis of claim 1, wherein the first end of the skirt-reinforcement member is circumferentially aligned with the second end of the skirt-reinforcement member such that the skirt-reinforcement member extends substantially perpendicular relative to an axis of the frame.

8. The prosthesis of claim 7, wherein the at least one side opening includes a maximum width and the skirt-reinforcement member spans across the at least one side opening at the maximum width.

9. The prosthesis of claim 7, wherein the frame includes a plurality of nodes, each node being a region where two of the plurality of crowns of the frame meet, and wherein the skirt-reinforcement member spans across the at least one side opening from one node to another node.

10. The prosthesis of claim 1, wherein the frame includes a row of side openings around a circumference of the frame and a skirt-reinforcement member spans each side opening of the row of side openings around the circumference of the frame.

11. The prosthesis of claim 1, wherein the frame is an inner frame and the prosthesis further comprises an outer frame coupled to the inner frame, the outer frame having a greater diameter than the inner frame.

12. The prosthesis of claim 1, wherein the prosthesis is a heart-valve prosthesis and the skirt-reinforcement member is positioned downstream of the margin of attachment.

13. The prosthesis of claim 12, wherein the heart-valve prosthesis is configured for placement within a mitral valve in situ.

14. A prosthesis having a radially expanded configuration and a radially compressed configuration, the prosthesis comprising:

a frame including a plurality of crowns and a plurality of struts with each crown being formed between a pair of

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opposing struts, wherein a plurality of side openings is defined by edges of the plurality of crowns and the plurality of struts;

a skirt coupled to a surface of the frame, wherein the skirt extends over at least one side opening of the plurality of side openings of the frame;

a prosthetic valve component including a plurality of leaflets disposed within and secured to the frame, the prosthetic valve being configured to block blood flow in one direction to regulate blood flow through a central lumen of the frame, wherein a margin of attachment is formed at the junction of the leaflets to the skirt; and

a skirt-reinforcement member that circumscribes an outer surface of the frame, wherein at least a portion of the skirt-reinforcement member is longitudinally spaced apart from the margin of attachment, wherein skirt-reinforcement member is attached to the skirt along an unsupported portion of the skirt that spans the at least one side opening of the plurality of side openings of the frame and wherein the skirt-reinforcement member is not directly attached to the frame.

15. The prosthesis of claim 14, wherein the skirt-reinforcement member is a ring of an elastic material.

16. The prosthesis of claim 14, wherein the at least one side opening of the plurality of side openings is substantially diamond-shaped.

17. The prosthesis of claim 14, wherein an end of the frame includes a row of side openings around a circumference of the frame, the row including between six and nine side openings.

18. The prosthesis of claim 14, wherein the frame is an inner frame and the prosthesis further comprises an outer frame coupled to the inner frame, the outer frame having a greater diameter than the inner frame.

19. The prosthesis of claim 14, wherein the prosthesis is a heart-valve prosthesis and the skirt-reinforcement member is positioned downstream of the margin of attachment.

20. The prosthesis of claim 19, wherein the heart-valve prosthesis is configured for placement within a mitral valve in situ.

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